Supplementary Information

Base denotes direct generation by the base model; Full denotes the complete ExpEvaFCHD workflow. Parentheses indicate 95% confidence intervals. For point estimates expressed as percentages, interval limits use the same percentage units. SD denotes standard deviation; TP, FP, TN and FN denote true positives, false positives, true negatives and false negatives, respectively.

Supplementary Table S1 | Quality metrics for generated reports stratified by reference diagnosis

Metrics are stratified by reference case labels. Means with 95% confidence intervals and SDs are reported separately. Confidence intervals for means were calculated using Student's t distribution.

Prospective cohort: negative cases

Model	Metric	n	Base · Direct generation		Full · ExpEvaFCHD	
			Mean (95% CI)	SD	Mean (95% CI)	SD
Kimi-K2	Cosine	1,118	0.838 (0.834, 0.842)	0.069	0.952 (0.951, 0.954)	0.033
Kimi-K2	CosineTI	1,118	0.558 (0.552, 0.564)	0.107	0.889 (0.886, 0.892)	0.050
Kimi-K2	Relevance	1,118	0.826 (0.821, 0.830)	0.078	0.974 (0.973, 0.976)	0.030
Kimi-K2	Informativeness	1,118	0.722 (0.716, 0.727)	0.098	0.365 (0.362, 0.368)	0.053
Kimi-K2	Validity	1,118	0.337 (0.332, 0.342)	0.080	0.511 (0.507, 0.514)	0.057
DeepSeek-V3.2	Cosine	1,118	0.830 (0.826, 0.835)	0.072	0.945 (0.943, 0.947)	0.034
DeepSeek-V3.2	CosineTI	1,118	0.553 (0.547, 0.560)	0.112	0.879 (0.876, 0.882)	0.051
DeepSeek-V3.2	Relevance	1,118	0.821 (0.817, 0.826)	0.080	0.967 (0.965, 0.969)	0.034
DeepSeek-V3.2	Informativeness	1,118	0.709 (0.703, 0.715)	0.099	0.358 (0.355, 0.362)	0.056
DeepSeek-V3.2	Validity	1,118	0.330 (0.325, 0.335)	0.079	0.503 (0.500, 0.507)	0.056
GLM-4.5	Cosine	1,118	0.836 (0.832, 0.840)	0.069	0.949 (0.947, 0.951)	0.033
GLM-4.5	CosineTI	1,118	0.557 (0.551, 0.564)	0.110	0.880 (0.877, 0.883)	0.050
GLM-4.5	Relevance	1,118	0.823 (0.819, 0.828)	0.079	0.971 (0.969, 0.973)	0.030
GLM-4.5	Informativeness	1,118	0.712 (0.707, 0.718)	0.100	0.362 (0.359, 0.365)	0.056
GLM-4.5	Validity	1,118	0.332 (0.328, 0.337)	0.081	0.505 (0.502, 0.509)	0.055
Llama-3.1-70B	Cosine	1,118	0.824 (0.819, 0.828)	0.069	0.939 (0.937, 0.941)	0.035
Llama-3.1-70B	CosineTI	1,118	0.545 (0.539, 0.551)	0.106	0.874 (0.871, 0.877)	0.051
Llama-3.1-70B	Relevance	1,118	0.817 (0.812, 0.821)	0.079	0.965 (0.963, 0.967)	0.032
Llama-3.1-70B	Informativeness	1,118	0.704 (0.698, 0.709)	0.098	0.352 (0.349, 0.356)	0.056
Llama-3.1-70B	Validity	1,118	0.325 (0.320, 0.330)	0.082	0.496 (0.492, 0.499)	0.057
Qwen3-Max	Cosine	1,118	0.842 (0.838, 0.847)	0.072	0.956 (0.954, 0.958)	0.033
Qwen3-Max	CosineTI	1,118	0.563 (0.556, 0.569)	0.115	0.891 (0.888, 0.894)	0.051
Qwen3-Max	Relevance	1,118	0.831 (0.826, 0.836)	0.080	0.977 (0.975, 0.978)	0.028
Qwen3-Max	Informativeness	1,118	0.725 (0.719, 0.731)	0.104	0.368 (0.364, 0.371)	0.058

Model	Metric	n	Base · Direct generation		Full · ExpEvaFCHD	
			Mean (95% CI)	SD	Mean (95% CI)	SD
Qwen3-Max	Validity	1,118	0.334 (0.330, 0.339)	0.083	0.513 (0.510, 0.517)	0.055
GPT-5	Cosine	1,118	0.844 (0.840, 0.848)	0.069	0.958 (0.956, 0.960)	0.032
GPT-5	CosineTI	1,118	0.567 (0.561, 0.574)	0.115	0.893 (0.890, 0.896)	0.049
GPT-5	Relevance	1,118	0.837 (0.833, 0.842)	0.080	0.978 (0.976, 0.980)	0.027
GPT-5	Informativeness	1,118	0.726 (0.720, 0.731)	0.096	0.372 (0.369, 0.375)	0.054
GPT-5	Validity	1,118	0.348 (0.344, 0.353)	0.078	0.517 (0.514, 0.521)	0.054

Prospective cohort: positive cases

Model	Metric	n	Base · Direct generation		Full · ExpEvaFCHD	
			Mean (95% CI)	SD	Mean (95% CI)	SD
Kimi-K2	Cosine	340	0.705 (0.697, 0.712)	0.073	0.879 (0.875, 0.883)	0.036
Kimi-K2	CosineTI	340	0.323 (0.312, 0.335)	0.106	0.750 (0.745, 0.756)	0.051
Kimi-K2	Relevance	340	0.664 (0.655, 0.673)	0.082	0.836 (0.832, 0.841)	0.042
Kimi-K2	Informativeness	340	0.785 (0.774, 0.795)	0.098	0.444 (0.438, 0.449)	0.053
Kimi-K2	Validity	340	0.284 (0.275, 0.292)	0.080	0.487 (0.481, 0.493)	0.057
DeepSeek-V3.2	Cosine	340	0.693 (0.685, 0.700)	0.069	0.867 (0.863, 0.871)	0.035
DeepSeek-V3.2	CosineTI	340	0.333 (0.321, 0.345)	0.115	0.749 (0.744, 0.755)	0.051
DeepSeek-V3.2	Relevance	340	0.645 (0.636, 0.654)	0.084	0.832 (0.827, 0.836)	0.040
DeepSeek-V3.2	Informativeness	340	0.789 (0.779, 0.799)	0.096	0.431 (0.425, 0.438)	0.058
DeepSeek-V3.2	Validity	340	0.269 (0.261, 0.277)	0.076	0.478 (0.472, 0.484)	0.057
GLM-4.5	Cosine	340	0.695 (0.688, 0.702)	0.065	0.871 (0.868, 0.875)	0.035
GLM-4.5	CosineTI	340	0.340 (0.328, 0.352)	0.113	0.754 (0.748, 0.760)	0.054
GLM-4.5	Relevance	340	0.654 (0.645, 0.662)	0.082	0.834 (0.829, 0.838)	0.040
GLM-4.5	Informativeness	340	0.788 (0.777, 0.798)	0.098	0.429 (0.423, 0.436)	0.060
GLM-4.5	Validity	340	0.270 (0.261, 0.279)	0.083	0.481 (0.475, 0.486)	0.054
Llama-3.1-70B	Cosine	340	0.694 (0.686, 0.702)	0.072	0.859 (0.856, 0.863)	0.036
Llama-3.1-70B	CosineTI	340	0.326 (0.314, 0.337)	0.110	0.735 (0.730, 0.741)	0.050
Llama-3.1-70B	Relevance	340	0.647 (0.638, 0.655)	0.082	0.826 (0.822, 0.830)	0.040
Llama-3.1-70B	Informativeness	340	0.777 (0.767, 0.788)	0.095	0.424 (0.418, 0.431)	0.059
Llama-3.1-70B	Validity	340	0.260 (0.252, 0.268)	0.076	0.469 (0.463, 0.475)	0.057
Qwen3-Max	Cosine	340	0.712 (0.705, 0.720)	0.071	0.877 (0.873, 0.881)	0.035
Qwen3-Max	CosineTI	340	0.337 (0.326, 0.347)	0.101	0.758 (0.752, 0.763)	0.051

Model	Metric	n	Base · Direct generation		Full · ExpEvaFCHD	
			Mean (95% CI)	SD	Mean (95% CI)	SD
Qwen3-Max	Relevance	340	0.664 (0.655, 0.672)	0.081	0.841 (0.837, 0.845)	0.041
Qwen3-Max	Informativeness	340	0.784 (0.774, 0.794)	0.096	0.441 (0.436, 0.447)	0.055
Qwen3-Max	Validity	340	0.284 (0.276, 0.292)	0.075	0.489 (0.483, 0.495)	0.054
GPT-5	Cosine	340	0.720 (0.712, 0.727)	0.069	0.884 (0.880, 0.887)	0.033
GPT-5	CosineTI	340	0.344 (0.332, 0.356)	0.113	0.763 (0.758, 0.769)	0.050
GPT-5	Relevance	340	0.664 (0.655, 0.672)	0.079	0.844 (0.840, 0.849)	0.043
GPT-5	Informativeness	340	0.794 (0.783, 0.805)	0.100	0.441 (0.435, 0.447)	0.053
GPT-5	Validity	340	0.280 (0.271, 0.289)	0.083	0.494 (0.488, 0.500)	0.056

Multicenter cohort: negative cases

Model	Metric	n	Base · Direct generation		Full · ExpEvaFCHD	
			Mean (95% CI)	SD	Mean (95% CI)	SD
Kimi-K2	Cosine	3,054	0.828 (0.826, 0.831)	0.070	0.944 (0.943, 0.945)	0.034
Kimi-K2	CosineTI	3,054	0.547 (0.543, 0.551)	0.109	0.877 (0.875, 0.879)	0.051
Kimi-K2	Relevance	3,054	0.820 (0.817, 0.823)	0.079	0.967 (0.966, 0.968)	0.032
Kimi-K2	Informativeness	3,054	0.706 (0.702, 0.710)	0.101	0.357 (0.355, 0.359)	0.053
Kimi-K2	Validity	3,054	0.329 (0.326, 0.332)	0.080	0.500 (0.498, 0.502)	0.056
DeepSeek-V3.2	Cosine	3,054	0.819 (0.817, 0.822)	0.071	0.934 (0.933, 0.935)	0.035
DeepSeek-V3.2	CosineTI	3,054	0.539 (0.535, 0.543)	0.111	0.867 (0.865, 0.868)	0.050
DeepSeek-V3.2	Relevance	3,054	0.810 (0.807, 0.812)	0.080	0.959 (0.957, 0.960)	0.034
DeepSeek-V3.2	Informativeness	3,054	0.701 (0.697, 0.704)	0.102	0.348 (0.346, 0.350)	0.055
DeepSeek-V3.2	Validity	3,054	0.317 (0.315, 0.320)	0.080	0.490 (0.488, 0.492)	0.056
GLM-4.5	Cosine	3,054	0.821 (0.818, 0.823)	0.069	0.937 (0.936, 0.938)	0.035
GLM-4.5	CosineTI	3,054	0.540 (0.537, 0.544)	0.109	0.871 (0.869, 0.873)	0.051
GLM-4.5	Relevance	3,054	0.812 (0.809, 0.815)	0.079	0.962 (0.961, 0.963)	0.034
GLM-4.5	Informativeness	3,054	0.702 (0.699, 0.706)	0.101	0.351 (0.349, 0.353)	0.055
GLM-4.5	Validity	3,054	0.321 (0.319, 0.324)	0.080	0.495 (0.493, 0.497)	0.055
Llama-3.1-70B	Cosine	3,054	0.813 (0.811, 0.816)	0.071	0.930 (0.929, 0.931)	0.036
Llama-3.1-70B	CosineTI	3,054	0.534 (0.530, 0.538)	0.114	0.863 (0.861, 0.864)	0.051
Llama-3.1-70B	Relevance	3,054	0.806 (0.803, 0.809)	0.080	0.954 (0.953, 0.956)	0.035
Llama-3.1-70B	Informativeness	3,054	0.693 (0.690, 0.697)	0.100	0.343 (0.341, 0.345)	0.056
Llama-3.1-70B	Validity	3,054	0.313 (0.310, 0.316)	0.082	0.486 (0.484, 0.488)	0.056

Model	Metric	n	Base · Direct generation		Full · ExpEvaFCHD	
			Mean (95% CI)	SD	Mean (95% CI)	SD
Qwen3-Max	Cosine	3,054	0.832 (0.830, 0.835)	0.070	0.945 (0.944, 0.947)	0.034
Qwen3-Max	CosineTI	3,054	0.549 (0.545, 0.553)	0.111	0.878 (0.876, 0.880)	0.050
Qwen3-Max	Relevance	3,054	0.818 (0.815, 0.820)	0.080	0.968 (0.967, 0.969)	0.032
Qwen3-Max	Informativeness	3,054	0.710 (0.706, 0.713)	0.099	0.360 (0.358, 0.362)	0.055
Qwen3-Max	Validity	3,054	0.330 (0.327, 0.333)	0.082	0.502 (0.500, 0.504)	0.056
GPT-5	Cosine	3,054	0.833 (0.830, 0.835)	0.070	0.948 (0.946, 0.949)	0.034
GPT-5	CosineTI	3,054	0.551 (0.548, 0.555)	0.108	0.882 (0.880, 0.884)	0.051
GPT-5	Relevance	3,054	0.823 (0.820, 0.826)	0.081	0.971 (0.970, 0.972)	0.031
GPT-5	Informativeness	3,054	0.712 (0.708, 0.715)	0.102	0.365 (0.363, 0.367)	0.056
GPT-5	Validity	3,054	0.334 (0.331, 0.337)	0.080	0.505 (0.503, 0.507)	0.057

Multicenter cohort: positive cases

Model	Metric	n	Base · Direct generation		Full · ExpEvaFCHD	
			Mean (95% CI)	SD	Mean (95% CI)	SD
Kimi-K2	Cosine	316	0.696 (0.688, 0.704)	0.074	0.867 (0.863, 0.871)	0.035
Kimi-K2	CosineTI	316	0.324 (0.311, 0.336)	0.113	0.740 (0.735, 0.745)	0.048
Kimi-K2	Relevance	316	0.646 (0.636, 0.655)	0.085	0.827 (0.823, 0.832)	0.040
Kimi-K2	Informativeness	316	0.773 (0.763, 0.784)	0.095	0.432 (0.426, 0.438)	0.054
Kimi-K2	Validity	316	0.270 (0.262, 0.279)	0.079	0.470 (0.464, 0.476)	0.054
DeepSeek-V3.2	Cosine	316	0.686 (0.678, 0.695)	0.075	0.856 (0.852, 0.860)	0.035
DeepSeek-V3.2	CosineTI	316	0.313 (0.301, 0.325)	0.109	0.736 (0.730, 0.741)	0.052
DeepSeek-V3.2	Relevance	316	0.629 (0.620, 0.637)	0.078	0.818 (0.814, 0.823)	0.041
DeepSeek-V3.2	Informativeness	316	0.772 (0.761, 0.784)	0.104	0.414 (0.408, 0.420)	0.055
DeepSeek-V3.2	Validity	316	0.253 (0.244, 0.262)	0.082	0.465 (0.459, 0.472)	0.055
GLM-4.5	Cosine	316	0.686 (0.678, 0.693)	0.069	0.862 (0.858, 0.866)	0.039
GLM-4.5	CosineTI	316	0.325 (0.314, 0.337)	0.106	0.732 (0.726, 0.738)	0.053
GLM-4.5	Relevance	316	0.646 (0.637, 0.655)	0.083	0.816 (0.811, 0.820)	0.041
GLM-4.5	Informativeness	316	0.776 (0.765, 0.786)	0.097	0.419 (0.413, 0.425)	0.055
GLM-4.5	Validity	316	0.262 (0.253, 0.271)	0.083	0.468 (0.462, 0.474)	0.054
Llama-3.1-70B	Cosine	316	0.680 (0.672, 0.688)	0.073	0.851 (0.847, 0.856)	0.037
Llama-3.1-70B	CosineTI	316	0.324 (0.311, 0.336)	0.115	0.732 (0.726, 0.738)	0.051
Llama-3.1-70B	Relevance	316	0.639 (0.630, 0.648)	0.081	0.812 (0.807, 0.816)	0.041

Model	Metric	n	Base · Direct generation		Full · ExpEvaFCHD	
			Mean (95% CI)	SD	Mean (95% CI)	SD
Llama-3.1-70B	Informativeness	316	0.754 (0.743, 0.764)	0.096	0.404 (0.398, 0.410)	0.055
Llama-3.1-70B	Validity	316	0.255 (0.246, 0.265)	0.082	0.459 (0.453, 0.465)	0.055
Qwen3-Max	Cosine	316	0.693 (0.685, 0.701)	0.072	0.870 (0.866, 0.874)	0.036
Qwen3-Max	CosineTI	316	0.321 (0.309, 0.333)	0.109	0.748 (0.742, 0.753)	0.051
Qwen3-Max	Relevance	316	0.650 (0.641, 0.658)	0.080	0.832 (0.827, 0.836)	0.041
Qwen3-Max	Informativeness	316	0.782 (0.772, 0.793)	0.095	0.432 (0.426, 0.438)	0.056
Qwen3-Max	Validity	316	0.267 (0.258, 0.276)	0.083	0.474 (0.468, 0.481)	0.057
GPT-5	Cosine	316	0.706 (0.698, 0.714)	0.073	0.872 (0.868, 0.876)	0.035
GPT-5	CosineTI	316	0.333 (0.320, 0.346)	0.116	0.753 (0.747, 0.759)	0.050
GPT-5	Relevance	316	0.657 (0.648, 0.666)	0.083	0.829 (0.825, 0.834)	0.042
GPT-5	Informativeness	316	0.790 (0.778, 0.801)	0.101	0.432 (0.425, 0.438)	0.057
GPT-5	Validity	316	0.275 (0.267, 0.283)	0.074	0.483 (0.476, 0.489)	0.056

Supplementary Table S2 | Paired differences in report quality between the full system and base models

Differences are Full minus Base. P (Wilcoxon) denotes the paired Wilcoxon test result; P (Holm) denotes the Holm-adjusted result.

Prospective cohort: negative cases

Model	Metric	n	Paired change	Significance tests	
			Full − Base (95% CI)	P (Wilcoxon)	P (Holm)
Kimi-K2	Cosine	1,118	0.115 (0.110, 0.119)	6.43E-173	2.19E-171
Kimi-K2	CosineTI	1,118	0.331 (0.324, 0.338)	2.09E-184	1.14E-182
Kimi-K2	Relevance	1,118	0.149 (0.144, 0.154)	3.92E-180	1.76E-178
Kimi-K2	Informativeness	1,118	-0.357 (-0.363, -0.350)	1.91E-184	1.14E-182
Kimi-K2	Validity	1,118	0.174 (0.168, 0.179)	7.23E-179	2.89E-177
DeepSeek-V3.2	Cosine	1,118	0.115 (0.110, 0.119)	5.50E-170	1.71E-168
DeepSeek-V3.2	CosineTI	1,118	0.326 (0.319, 0.333)	2.46E-184	1.23E-182
DeepSeek-V3.2	Relevance	1,118	0.146 (0.141, 0.151)	3.25E-178	1.20E-176
DeepSeek-V3.2	Informativeness	1,118	-0.351 (-0.358, -0.344)	1.94E-184	1.14E-182
DeepSeek-V3.2	Validity	1,118	0.173 (0.168, 0.179)	2.62E-181	1.26E-179
GLM-4.5	Cosine	1,118	0.113 (0.108, 0.117)	2.73E-173	9.57E-172
GLM-4.5	CosineTI	1,118	0.323 (0.316, 0.330)	2.01E-184	1.14E-182
GLM-4.5	Relevance	1,118	0.148 (0.143, 0.152)	7.46E-180	3.28E-178
GLM-4.5	Informativeness	1,118	-0.350 (-0.357, -0.344)	1.93E-184	1.14E-182

Model	Metric	n	Paired change	Significance tests	
			Full − Base (95% CI)	P (Wilcoxon)	P (Holm)
GLM-4.5	Validity	1,118	0.173 (0.167, 0.179)	1.35E-179	5.82E-178
Llama-3.1-70B	Cosine	1,118	0.115 (0.111, 0.120)	8.59E-173	2.83E-171
Llama-3.1-70B	CosineTI	1,118	0.329 (0.322, 0.336)	1.91E-184	1.14E-182
Llama-3.1-70B	Relevance	1,118	0.148 (0.143, 0.153)	2.92E-179	1.23E-177
Llama-3.1-70B	Informativeness	1,118	-0.351 (-0.358, -0.345)	1.92E-184	1.14E-182
Llama-3.1-70B	Validity	1,118	0.171 (0.165, 0.177)	4.12E-179	1.69E-177
Qwen3-Max	Cosine	1,118	0.113 (0.109, 0.118)	6.83E-171	2.19E-169
Qwen3-Max	CosineTI	1,118	0.328 (0.321, 0.335)	2.19E-184	1.14E-182
Qwen3-Max	Relevance	1,118	0.146 (0.141, 0.150)	1.29E-178	4.89E-177
Qwen3-Max	Informativeness	1,118	-0.357 (-0.364, -0.350)	1.91E-184	1.14E-182
Qwen3-Max	Validity	1,118	0.179 (0.173, 0.185)	1.45E-180	6.67E-179
GPT-5	Cosine	1,118	0.114 (0.110, 0.119)	1.67E-173	6.03E-172
GPT-5	CosineTI	1,118	0.326 (0.318, 0.333)	3.75E-184	1.84E-182
GPT-5	Relevance	1,118	0.141 (0.136, 0.145)	8.68E-179	3.39E-177
GPT-5	Informativeness	1,118	-0.354 (-0.360, -0.347)	1.91E-184	1.14E-182
GPT-5	Validity	1,118	0.169 (0.163, 0.174)	5.78E-181	2.72E-179

Prospective cohort: positive cases

Model	Metric	n	Paired change	Significance tests	
			Full − Base (95% CI)	P (Wilcoxon)	P (Holm)
Kimi-K2	Cosine	340	0.175 (0.166, 0.183)	2.05E-57	5.25E-56
Kimi-K2	CosineTI	340	0.427 (0.415, 0.439)	1.75E-57	5.25E-56
Kimi-K2	Relevance	340	0.173 (0.163, 0.182)	1.61E-56	5.25E-56
Kimi-K2	Informativeness	340	-0.341 (-0.353, -0.330)	1.78E-57	5.25E-56
Kimi-K2	Validity	340	0.203 (0.192, 0.214)	5.04E-57	5.25E-56
DeepSeek-V3.2	Cosine	340	0.174 (0.166, 0.182)	1.86E-57	5.25E-56
DeepSeek-V3.2	CosineTI	340	0.416 (0.403, 0.429)	1.75E-57	5.25E-56
DeepSeek-V3.2	Relevance	340	0.187 (0.177, 0.197)	4.01E-57	5.25E-56
DeepSeek-V3.2	Informativeness	340	-0.358 (-0.370, -0.346)	1.75E-57	5.25E-56
DeepSeek-V3.2	Validity	340	0.209 (0.199, 0.218)	2.32E-57	5.25E-56
GLM-4.5	Cosine	340	0.176 (0.168, 0.184)	2.95E-57	5.25E-56
GLM-4.5	CosineTI	340	0.414 (0.401, 0.428)	1.75E-57	5.25E-56

Model	Metric	n	Paired change	Significance tests	
			Full − Base (95% CI)	P (Wilcoxon)	P (Holm)
GLM-4.5	Relevance	340	0.180 (0.171, 0.189)	5.13E-57	5.25E-56
GLM-4.5	Informativeness	340	-0.358 (-0.370, -0.346)	1.75E-57	5.25E-56
GLM-4.5	Validity	340	0.210 (0.200, 0.221)	3.58E-57	5.25E-56
Llama-3.1-70B	Cosine	340	0.166 (0.157, 0.174)	4.46E-57	5.25E-56
Llama-3.1-70B	CosineTI	340	0.410 (0.397, 0.423)	1.75E-57	5.25E-56
Llama-3.1-70B	Relevance	340	0.179 (0.170, 0.189)	1.10E-55	1.10E-55
Llama-3.1-70B	Informativeness	340	-0.353 (-0.365, -0.341)	1.75E-57	5.25E-56
Llama-3.1-70B	Validity	340	0.209 (0.199, 0.220)	2.67E-57	5.25E-56
Qwen3-Max	Cosine	340	0.165 (0.156, 0.173)	5.37E-57	5.25E-56
Qwen3-Max	CosineTI	340	0.421 (0.409, 0.433)	1.75E-57	5.25E-56
Qwen3-Max	Relevance	340	0.177 (0.168, 0.187)	7.56E-57	5.25E-56
Qwen3-Max	Informativeness	340	-0.343 (-0.354, -0.331)	1.75E-57	5.25E-56
Qwen3-Max	Validity	340	0.205 (0.196, 0.214)	1.93E-57	5.25E-56
GPT-5	Cosine	340	0.164 (0.156, 0.172)	2.24E-57	5.25E-56
GPT-5	CosineTI	340	0.420 (0.406, 0.433)	1.76E-57	5.25E-56
GPT-5	Relevance	340	0.181 (0.171, 0.190)	5.81E-57	5.25E-56
GPT-5	Informativeness	340	-0.353 (-0.365, -0.341)	1.78E-57	5.25E-56
GPT-5	Validity	340	0.214 (0.203, 0.225)	5.71E-57	5.25E-56

Multicenter cohort: negative cases

Model	Metric	n	Paired change	Significance tests	
			Full − Base (95% CI)	P (Wilcoxon)	P (Holm)
Kimi-K2	Cosine	3,054	0.116 (0.113, 0.118)	<1e-300	<1e-300
Kimi-K2	CosineTI	3,054	0.330 (0.326, 0.334)	<1e-300	<1e-300
Kimi-K2	Relevance	3,054	0.147 (0.144, 0.150)	<1e-300	<1e-300
Kimi-K2	Informativeness	3,054	-0.349 (-0.353, -0.345)	<1e-300	<1e-300
Kimi-K2	Validity	3,054	0.171 (0.168, 0.174)	<1e-300	<1e-300
DeepSeek-V3.2	Cosine	3,054	0.115 (0.112, 0.118)	<1e-300	<1e-300
DeepSeek-V3.2	CosineTI	3,054	0.328 (0.324, 0.332)	<1e-300	<1e-300
DeepSeek-V3.2	Relevance	3,054	0.149 (0.146, 0.152)	<1e-300	<1e-300
DeepSeek-V3.2	Informativeness	3,054	-0.353 (-0.357, -0.349)	<1e-300	<1e-300
DeepSeek-V3.2	Validity	3,054	0.172 (0.169, 0.176)	<1e-300	<1e-300

Model	Metric	n	Paired change	Significance tests	
			Full − Base (95% CI)	P (Wilcoxon)	P (Holm)
GLM-4.5	Cosine	3,054	0.116 (0.113, 0.119)	<1e-300	<1e-300
GLM-4.5	CosineTI	3,054	0.330 (0.326, 0.335)	<1e-300	<1e-300
GLM-4.5	Relevance	3,054	0.150 (0.147, 0.153)	<1e-300	<1e-300
GLM-4.5	Informativeness	3,054	-0.352 (-0.356, -0.348)	<1e-300	<1e-300
GLM-4.5	Validity	3,054	0.173 (0.170, 0.177)	<1e-300	<1e-300
Llama-3.1-70B	Cosine	3,054	0.117 (0.114, 0.119)	<1e-300	<1e-300
Llama-3.1-70B	CosineTI	3,054	0.329 (0.325, 0.333)	<1e-300	<1e-300
Llama-3.1-70B	Relevance	3,054	0.149 (0.146, 0.152)	<1e-300	<1e-300
Llama-3.1-70B	Informativeness	3,054	-0.351 (-0.355, -0.347)	<1e-300	<1e-300
Llama-3.1-70B	Validity	3,054	0.172 (0.169, 0.176)	<1e-300	<1e-300
Qwen3-Max	Cosine	3,054	0.113 (0.111, 0.116)	<1e-300	<1e-300
Qwen3-Max	CosineTI	3,054	0.329 (0.325, 0.334)	<1e-300	<1e-300
Qwen3-Max	Relevance	3,054	0.150 (0.147, 0.153)	<1e-300	<1e-300
Qwen3-Max	Informativeness	3,054	-0.350 (-0.354, -0.346)	<1e-300	<1e-300
Qwen3-Max	Validity	3,054	0.172 (0.168, 0.175)	<1e-300	<1e-300
GPT-5	Cosine	3,054	0.115 (0.112, 0.118)	<1e-300	<1e-300
GPT-5	CosineTI	3,054	0.331 (0.326, 0.335)	<1e-300	<1e-300
GPT-5	Relevance	3,054	0.147 (0.144, 0.151)	<1e-300	<1e-300
GPT-5	Informativeness	3,054	-0.347 (-0.351, -0.343)	<1e-300	<1e-300
GPT-5	Validity	3,054	0.171 (0.168, 0.174)	<1e-300	<1e-300

Multicenter cohort: positive cases

Model	Metric	n	Paired change	Significance tests	
			Full − Base (95% CI)	P (Wilcoxon)	P (Holm)
Kimi-K2	Cosine	316	0.171 (0.162, 0.180)	5.34E-53	4.41E-52
Kimi-K2	CosineTI	316	0.416 (0.403, 0.430)	1.51E-53	4.41E-52
Kimi-K2	Relevance	316	0.181 (0.171, 0.192)	4.86E-53	4.41E-52
Kimi-K2	Informativeness	316	-0.341 (-0.353, -0.329)	1.47E-53	4.41E-52
Kimi-K2	Validity	316	0.200 (0.188, 0.211)	3.91E-53	4.41E-52
DeepSeek-V3.2	Cosine	316	0.170 (0.161, 0.179)	2.50E-53	4.41E-52
DeepSeek-V3.2	CosineTI	316	0.423 (0.409, 0.436)	1.47E-53	4.41E-52
DeepSeek-V3.2	Relevance	316	0.190 (0.180, 0.199)	2.62E-53	4.41E-52

Model	Metric	n	Paired change	Significance tests	
			Full − Base (95% CI)	P (Wilcoxon)	P (Holm)
DeepSeek-V3.2	Informativeness	316	-0.358 (-0.371, -0.345)	1.47E-53	4.41E-52
DeepSeek-V3.2	Validity	316	0.213 (0.202, 0.223)	1.62E-53	4.41E-52
GLM-4.5	Cosine	316	0.177 (0.168, 0.185)	2.03E-53	4.41E-52
GLM-4.5	CosineTI	316	0.407 (0.393, 0.420)	1.47E-53	4.41E-52
GLM-4.5	Relevance	316	0.170 (0.160, 0.180)	8.99E-53	4.41E-52
GLM-4.5	Informativeness	316	-0.357 (-0.369, -0.344)	1.47E-53	4.41E-52
GLM-4.5	Validity	316	0.206 (0.194, 0.217)	3.03E-53	4.41E-52
Llama-3.1-70B	Cosine	316	0.172 (0.163, 0.181)	2.65E-53	4.41E-52
Llama-3.1-70B	CosineTI	316	0.408 (0.394, 0.422)	1.47E-53	4.41E-52
Llama-3.1-70B	Relevance	316	0.173 (0.163, 0.183)	7.80E-53	4.41E-52
Llama-3.1-70B	Informativeness	316	-0.350 (-0.362, -0.338)	1.47E-53	4.41E-52
Llama-3.1-70B	Validity	316	0.204 (0.192, 0.215)	3.33E-53	4.41E-52
Qwen3-Max	Cosine	316	0.177 (0.168, 0.185)	1.76E-53	4.41E-52
Qwen3-Max	CosineTI	316	0.426 (0.413, 0.439)	1.47E-53	4.41E-52
Qwen3-Max	Relevance	316	0.182 (0.172, 0.192)	1.25E-52	4.41E-52
Qwen3-Max	Informativeness	316	-0.350 (-0.362, -0.339)	1.47E-53	4.41E-52
Qwen3-Max	Validity	316	0.207 (0.196, 0.218)	2.60E-53	4.41E-52
GPT-5	Cosine	316	0.166 (0.157, 0.175)	2.12E-52	4.41E-52
GPT-5	CosineTI	316	0.420 (0.406, 0.434)	1.47E-53	4.41E-52
GPT-5	Relevance	316	0.172 (0.162, 0.183)	1.40E-52	4.41E-52
GPT-5	Informativeness	316	-0.358 (-0.371, -0.345)	1.47E-53	4.41E-52
GPT-5	Validity	316	0.207 (0.198, 0.217)	1.78E-53	4.41E-52

Supplementary Table S3 | Recognition performance across seven fetal cardiac disease families

Each disease family is evaluated separately as the positive target. Reference-positive counts indicate the number of cases in that family. Counts are shown in the order TP / FP / TN / FN. A case may belong to more than one family. A dash indicates that recall could not be estimated because there were 0 reference-positive cases.

Prospective cohort: Base

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
Kimi-K2	Septal defects	139	29 / 11 / 1,308 / 110	20.86% (14.94, 28.36)	99.17% (98.51, 99.53)	72.50% (57.17, 83.89)
Kimi-K2	Great vessels/aortic arch	116	16 / 17 / 1,325 / 100	13.79% (8.67, 21.24)	98.73% (97.98, 99.21)	48.48% (32.50, 64.78)
Kimi-K2	Venous return/systemic veins	47	5 / 22 / 1,389 / 42	10.64% (4.63, 22.59)	98.44% (97.65, 98.97)	18.52% (8.18, 36.70)

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
Kimi-K2	Valves/outflow tracts	77	12 / 21 / 1,360 / 65	15.58% (9.15, 25.29)	98.48% (97.69, 99.00)	36.36% (22.19, 53.38)
Kimi-K2	Physiological shunt abnormalities	63	10 / 22 / 1,373 / 53	15.87% (8.86, 26.81)	98.42% (97.62, 98.96)	31.25% (17.95, 48.57)
Kimi-K2	Other structural abnormalities	53	9 / 19 / 1,386 / 44	16.98% (9.20, 29.23)	98.65% (97.90, 99.13)	32.14% (17.93, 50.66)
Kimi-K2	Cardiac position abnormalities	0	0 / 20 / 1,438 / 0	-	98.63% (97.89, 99.11)	0.00% (0.00, 16.11)
DeepSeek-V3.2	Septal defects	139	22 / 43 / 1,276 / 117	15.83% (10.69, 22.80)	96.74% (95.64, 97.57)	33.85% (23.53, 45.96)
DeepSeek-V3.2	Great vessels/aortic arch	116	10 / 48 / 1,294 / 106	8.62% (4.75, 15.14)	96.42% (95.29, 97.29)	17.24% (9.64, 28.91)
DeepSeek-V3.2	Venous return/systemic veins	47	3 / 47 / 1,364 / 44	6.38% (2.19, 17.16)	96.67% (95.60, 97.49)	6.00% (2.06, 16.22)
DeepSeek-V3.2	Valves/outflow tracts	77	7 / 47 / 1,334 / 70	9.09% (4.47, 17.60)	96.60% (95.50, 97.43)	12.96% (6.42, 24.42)
DeepSeek-V3.2	Physiological shunt abnormalities	63	7 / 47 / 1,348 / 56	11.11% (5.49, 21.20)	96.63% (95.55, 97.46)	12.96% (6.42, 24.42)
DeepSeek-V3.2	Other structural abnormalities	53	6 / 44 / 1,361 / 47	11.32% (5.29, 22.58)	96.87% (95.82, 97.66)	12.00% (5.62, 23.80)
DeepSeek-V3.2	Cardiac position abnormalities	0	0 / 53 / 1,405 / 0	-	96.36% (95.28, 97.21)	0.00% (0.00, 6.76)
GLM-4.5	Septal defects	139	22 / 21 / 1,298 / 117	15.83% (10.69, 22.80)	98.41% (97.58, 98.96)	51.16% (36.75, 65.38)
GLM-4.5	Great vessels/aortic arch	116	10 / 14 / 1,328 / 106	8.62% (4.75, 15.14)	98.96% (98.26, 99.38)	41.67% (24.47, 61.17)
GLM-4.5	Venous return/systemic veins	47	3 / 23 / 1,388 / 44	6.38% (2.19, 17.16)	98.37% (97.57, 98.91)	11.54% (4.00, 28.98)
GLM-4.5	Valves/outflow tracts	77	8 / 22 / 1,359 / 69	10.39% (5.36, 19.18)	98.41% (97.60, 98.95)	26.67% (14.18, 44.45)
GLM-4.5	Physiological shunt abnormalities	63	7 / 28 / 1,367 / 56	11.11% (5.49, 21.20)	97.99% (97.11, 98.61)	20.00% (10.04, 35.89)
GLM-4.5	Other structural abnormalities	53	6 / 10 / 1,395 / 47	11.32% (5.29, 22.58)	99.29% (98.69, 99.61)	37.50% (18.48, 61.36)
GLM-4.5	Cardiac position abnormalities	0	0 / 19 / 1,439 / 0	-	98.70% (97.97, 99.16)	0.00% (0.00, 16.82)
Llama-3.1-70B	Septal defects	139	19 / 19 / 1,300 / 120	13.67% (8.93, 20.36)	98.56% (97.76, 99.08)	50.00% (34.85, 65.15)
Llama-3.1-70B	Great vessels/aortic arch	116	8 / 14 / 1,328 / 108	6.90% (3.54, 13.02)	98.96% (98.26, 99.38)	36.36% (19.73, 57.05)
Llama-3.1-70B	Venous return/systemic veins	47	2 / 21 / 1,390 / 45	4.26% (1.17, 14.25)	98.51% (97.74, 99.02)	8.70% (2.42, 26.80)
Llama-3.1-70B	Valves/outflow tracts	77	6 / 23 / 1,358 / 71	7.79% (3.62, 15.98)	98.33% (97.51, 98.89)	20.69% (9.85, 38.39)
Llama-3.1-70B	Physiological shunt abnormalities	63	6 / 19 / 1,376 / 57	9.52% (4.44, 19.26)	98.64% (97.88, 99.13)	24.00% (11.50, 43.43)
Llama-3.1-70B	Other structural abnormalities	53	5 / 19 / 1,386 / 48	9.43% (4.10, 20.25)	98.65% (97.90, 99.13)	20.83% (9.24, 40.47)
Llama-3.1-70B	Cardiac position abnormalities	0	0 / 20 / 1,438 / 0	-	98.63% (97.89, 99.11)	0.00% (0.00, 16.11)
Qwen3-Max	Septal defects	139	25 / 17 / 1,302 / 114	17.99% (12.49, 25.20)	98.71% (97.95, 99.19)	59.52% (44.49, 72.96)
Qwen3-Max	Great vessels/aortic arch	116	13 / 10 / 1,332 / 103	11.21% (6.67, 18.23)	99.25% (98.63, 99.59)	56.52% (36.81, 74.37)
Qwen3-Max	Venous return/systemic veins	47	4 / 18 / 1,393 / 43	8.51% (3.36, 19.93)	98.72% (97.99, 99.19)	18.18% (7.31, 38.52)
Qwen3-Max	Valves/outflow tracts	77	9 / 15 / 1,366 / 68	11.69% (6.27, 20.75)	98.91% (98.22, 99.34)	37.50% (21.16, 57.29)
Qwen3-Max	Physiological shunt abnormalities	63	8 / 14 / 1,381 / 55	12.70% (6.58, 23.11)	99.00% (98.32, 99.40)	36.36% (19.73, 57.05)
Qwen3-Max	Other structural abnormalities	53	7 / 16 / 1,389 / 46	13.21% (6.55, 24.84)	98.86% (98.16, 99.30)	30.43% (15.60, 50.87)
Qwen3-Max	Cardiac position abnormalities	0	0 / 25 / 1,433 / 0	-	98.29% (97.48, 98.84)	0.00% (0.00, 13.32)

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
GPT-5	Septal defects	139	28 / 23 / 1,296 / 111	20.14% (14.32, 27.57)	98.26% (97.40, 98.84)	54.90% (41.38, 67.73)
GPT-5	Great vessels/aortic arch	116	15 / 15 / 1,327 / 101	12.93% (8.00, 20.24)	98.88% (98.16, 99.32)	50.00% (33.15, 66.85)
GPT-5	Venous return/systemic veins	47	5 / 16 / 1,395 / 42	10.64% (4.63, 22.59)	98.87% (98.17, 99.30)	23.81% (10.63, 45.09)
GPT-5	Valves/outflow tracts	77	11 / 18 / 1,363 / 66	14.29% (8.17, 23.80)	98.70% (97.95, 99.17)	37.93% (22.69, 56.00)
GPT-5	Physiological shunt abnormalities	63	9 / 15 / 1,380 / 54	14.29% (7.70, 24.97)	98.92% (98.23, 99.35)	37.50% (21.16, 57.29)
GPT-5	Other structural abnormalities	53	8 / 23 / 1,382 / 45	15.09% (7.85, 27.05)	98.36% (97.56, 98.91)	25.81% (13.70, 43.25)
GPT-5	Cardiac position abnormalities	0	0 / 21 / 1,437 / 0	-	98.56% (97.81, 99.06)	0.00% (0.00, 15.46)

Prospective cohort: Full

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
Kimi-K2	Septal defects	139	108 / 8 / 1,311 / 31	77.70% (70.09, 83.82)	99.39% (98.81, 99.69)	93.10% (86.98, 96.46)
Kimi-K2	Great vessels/aortic arch	116	71 / 4 / 1,338 / 45	61.21% (52.12, 69.58)	99.70% (99.24, 99.88)	94.67% (87.07, 97.91)
Kimi-K2	Venous return/systemic veins	47	20 / 5 / 1,406 / 27	42.55% (29.51, 56.72)	99.65% (99.17, 99.85)	80.00% (60.87, 91.14)
Kimi-K2	Valves/outflow tracts	77	49 / 7 / 1,374 / 28	63.64% (52.48, 73.49)	99.49% (98.96, 99.75)	87.50% (76.37, 93.81)
Kimi-K2	Physiological shunt abnormalities	63	33 / 6 / 1,389 / 30	52.38% (40.27, 64.22)	99.57% (99.06, 99.80)	84.62% (70.27, 92.75)
Kimi-K2	Other structural abnormalities	53	31 / 7 / 1,398 / 22	58.49% (45.09, 70.74)	99.50% (98.98, 99.76)	81.58% (66.58, 90.78)
Kimi-K2	Cardiac position abnormalities	0	0 / 4 / 1,454 / 0	-	99.73% (99.30, 99.89)	0.00% (0.00, 48.99)
DeepSeek-V3.2	Septal defects	139	101 / 4 / 1,315 / 38	72.66% (64.72, 79.39)	99.70% (99.22, 99.88)	96.19% (90.61, 98.51)
DeepSeek-V3.2	Great vessels/aortic arch	116	64 / 3 / 1,339 / 52	55.17% (46.10, 63.91)	99.78% (99.34, 99.92)	95.52% (87.64, 98.47)
DeepSeek-V3.2	Venous return/systemic veins	47	18 / 6 / 1,405 / 29	38.30% (25.79, 52.57)	99.57% (99.08, 99.80)	75.00% (55.10, 88.00)
DeepSeek-V3.2	Valves/outflow tracts	77	44 / 5 / 1,376 / 33	57.14% (46.01, 67.60)	99.64% (99.16, 99.85)	89.80% (78.24, 95.56)
DeepSeek-V3.2	Physiological shunt abnormalities	63	30 / 8 / 1,387 / 33	47.62% (35.78, 59.73)	99.43% (98.87, 99.71)	78.95% (63.65, 88.93)
DeepSeek-V3.2	Other structural abnormalities	53	28 / 6 / 1,399 / 25	52.83% (39.66, 65.62)	99.57% (99.07, 99.80)	82.35% (66.49, 91.65)
DeepSeek-V3.2	Cardiac position abnormalities	0	0 / 4 / 1,454 / 0	-	99.73% (99.30, 99.89)	0.00% (0.00, 48.99)
GLM-4.5	Septal defects	139	101 / 3 / 1,316 / 38	72.66% (64.72, 79.39)	99.77% (99.33, 99.92)	97.12% (91.86, 99.01)
GLM-4.5	Great vessels/aortic arch	116	65 / 7 / 1,335 / 51	56.03% (46.95, 64.73)	99.48% (98.93, 99.75)	90.28% (81.26, 95.21)
GLM-4.5	Venous return/systemic veins	47	18 / 5 / 1,406 / 29	38.30% (25.79, 52.57)	99.65% (99.17, 99.85)	78.26% (58.10, 90.34)
GLM-4.5	Valves/outflow tracts	77	45 / 8 / 1,373 / 32	58.44% (47.29, 68.79)	99.42% (98.86, 99.71)	84.91% (72.95, 92.15)
GLM-4.5	Physiological shunt abnormalities	63	30 / 4 / 1,391 / 33	47.62% (35.78, 59.73)	99.71% (99.27, 99.89)	88.24% (73.38, 95.33)
GLM-4.5	Other structural abnormalities	53	28 / 6 / 1,399 / 25	52.83% (39.66, 65.62)	99.57% (99.07, 99.80)	82.35% (66.49, 91.65)
GLM-4.5	Cardiac position abnormalities	0	0 / 7 / 1,451 / 0	-	99.52% (99.01, 99.77)	0.00% (0.00, 35.43)

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
Llama-3.1-70B	Septal defects	139	99 / 3 / 1,316 / 40	71.22% (63.21, 78.10)	99.77% (99.33, 99.92)	97.06% (91.71, 98.99)
Llama-3.1-70B	Great vessels/aortic arch	116	63 / 6 / 1,336 / 53	54.31% (45.25, 63.09)	99.55% (99.03, 99.79)	91.30% (82.30, 95.95)
Llama-3.1-70B	Venous return/systemic veins	47	17 / 3 / 1,408 / 30	36.17% (23.97, 50.46)	99.79% (99.38, 99.93)	85.00% (63.96, 94.76)
Llama-3.1-70B	Valves/outflow tracts	77	43 / 2 / 1,379 / 34	55.84% (44.74, 66.39)	99.86% (99.47, 99.96)	95.56% (85.17, 98.77)
Llama-3.1-70B	Physiological shunt abnormalities	63	29 / 3 / 1,392 / 34	46.03% (34.31, 58.21)	99.78% (99.37, 99.93)	90.63% (75.78, 96.76)
Llama-3.1-70B	Other structural abnormalities	53	27 / 10 / 1,395 / 26	50.94% (37.88, 63.88)	99.29% (98.69, 99.61)	72.97% (57.02, 84.60)
Llama-3.1-70B	Cardiac position abnormalities	0	0 / 6 / 1,452 / 0	-	99.59% (99.11, 99.81)	0.00% (0.00, 39.03)
Qwen3-Max	Septal defects	139	104 / 2 / 1,317 / 35	74.82% (67.00, 81.30)	99.85% (99.45, 99.96)	98.11% (93.38, 99.48)
Qwen3-Max	Great vessels/aortic arch	116	67 / 3 / 1,339 / 49	57.76% (48.66, 66.36)	99.78% (99.34, 99.92)	95.71% (88.14, 98.53)
Qwen3-Max	Venous return/systemic veins	47	19 / 5 / 1,406 / 28	40.43% (27.64, 54.66)	99.65% (99.17, 99.85)	79.17% (59.53, 90.76)
Qwen3-Max	Valves/outflow tracts	77	46 / 4 / 1,377 / 31	59.74% (48.58, 69.98)	99.71% (99.26, 99.89)	92.00% (81.16, 96.85)
Qwen3-Max	Physiological shunt abnormalities	63	32 / 6 / 1,389 / 31	50.79% (38.76, 62.73)	99.57% (99.06, 99.80)	84.21% (69.58, 92.56)
Qwen3-Max	Other structural abnormalities	53	29 / 3 / 1,402 / 24	54.72% (41.45, 67.34)	99.79% (99.37, 99.93)	90.63% (75.78, 96.76)
Qwen3-Max	Cardiac position abnormalities	0	0 / 3 / 1,455 / 0	-	99.79% (99.40, 99.93)	0.00% (0.00, 56.15)
GPT-5	Septal defects	139	107 / 5 / 1,314 / 32	76.98% (69.31, 83.19)	99.62% (99.12, 99.84)	95.54% (89.97, 98.08)
GPT-5	Great vessels/aortic arch	116	70 / 4 / 1,338 / 46	60.34% (51.25, 68.78)	99.70% (99.24, 99.88)	94.59% (86.91, 97.88)
GPT-5	Venous return/systemic veins	47	20 / 6 / 1,405 / 27	42.55% (29.51, 56.72)	99.57% (99.08, 99.80)	76.92% (57.95, 88.97)
GPT-5	Valves/outflow tracts	77	48 / 8 / 1,373 / 29	62.34% (51.17, 72.33)	99.42% (98.86, 99.71)	85.71% (74.26, 92.58)
GPT-5	Physiological shunt abnormalities	63	33 / 7 / 1,388 / 30	52.38% (40.27, 64.22)	99.50% (98.97, 99.76)	82.50% (68.05, 91.25)
GPT-5	Other structural abnormalities	53	30 / 3 / 1,402 / 23	56.60% (43.27, 69.05)	99.79% (99.37, 99.93)	90.91% (76.43, 96.86)
GPT-5	Cardiac position abnormalities	0	0 / 7 / 1,451 / 0	-	99.52% (99.01, 99.77)	0.00% (0.00, 35.43)

Multicenter cohort: Base

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
Kimi-K2	Septal defects	139	25 / 34 / 3,197 / 114	17.99% (12.49, 25.20)	98.95% (98.53, 99.25)	42.37% (30.61, 55.07)
Kimi-K2	Great vessels/aortic arch	123	14 / 37 / 3,210 / 109	11.38% (6.90, 18.20)	98.86% (98.43, 99.17)	27.45% (17.11, 40.95)
Kimi-K2	Venous return/systemic veins	34	3 / 28 / 3,308 / 31	8.82% (3.05, 22.96)	99.16% (98.79, 99.42)	9.68% (3.35, 24.90)
Kimi-K2	Valves/outflow tracts	45	5 / 37 / 3,288 / 40	11.11% (4.84, 23.50)	98.89% (98.47, 99.19)	11.90% (5.19, 25.00)
Kimi-K2	Physiological shunt abnormalities	31	4 / 29 / 3,310 / 27	12.90% (5.13, 28.85)	99.13% (98.76, 99.39)	12.12% (4.82, 27.33)
Kimi-K2	Other structural abnormalities	56	8 / 38 / 3,276 / 48	14.29% (7.42, 25.74)	98.85% (98.43, 99.16)	17.39% (9.09, 30.72)
Kimi-K2	Cardiac position abnormalities	2	0 / 40 / 3,328 / 2	0.00% (0.00, 65.76)	98.81% (98.39, 99.13)	0.00% (0.00, 8.76)

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
DeepSeek-V3.2	Septal defects	139	17 / 127 / 3,104 / 122	12.23% (7.78, 18.71)	96.07% (95.34, 96.69)	11.81% (7.50, 18.09)
DeepSeek-V3.2	Great vessels/aortic arch	123	7 / 119 / 3,128 / 116	5.69% (2.78, 11.28)	96.34% (95.63, 96.93)	5.56% (2.72, 11.02)
DeepSeek-V3.2	Venous return/systemic veins	34	1 / 128 / 3,208 / 33	2.94% (0.52, 14.92)	96.16% (95.46, 96.76)	0.78% (0.14, 4.26)
DeepSeek-V3.2	Valves/outflow tracts	45	3 / 118 / 3,207 / 42	6.67% (2.29, 17.86)	96.45% (95.77, 97.03)	2.48% (0.85, 7.04)
DeepSeek-V3.2	Physiological shunt abnormalities	31	2 / 118 / 3,221 / 29	6.45% (1.79, 20.72)	96.47% (95.78, 97.04)	1.67% (0.46, 5.87)
DeepSeek-V3.2	Other structural abnormalities	56	5 / 129 / 3,185 / 51	8.93% (3.87, 19.26)	96.11% (95.39, 96.71)	3.73% (1.60, 8.44)
DeepSeek-V3.2	Cardiac position abnormalities	2	0 / 122 / 3,246 / 2	0.00% (0.00, 65.76)	96.38% (95.69, 96.96)	0.00% (0.00, 3.05)
GLM-4.5	Septal defects	139	18 / 38 / 3,193 / 121	12.95% (8.35, 19.54)	98.82% (98.39, 99.14)	32.14% (21.40, 45.18)
GLM-4.5	Great vessels/aortic arch	123	7 / 28 / 3,219 / 116	5.69% (2.78, 11.28)	99.14% (98.76, 99.40)	20.00% (10.04, 35.89)
GLM-4.5	Venous return/systemic veins	34	1 / 49 / 3,287 / 33	2.94% (0.52, 14.92)	98.53% (98.06, 98.89)	2.00% (0.35, 10.50)
GLM-4.5	Valves/outflow tracts	45	3 / 42 / 3,283 / 42	6.67% (2.29, 17.86)	98.74% (98.30, 99.06)	6.67% (2.29, 17.86)
GLM-4.5	Physiological shunt abnormalities	31	2 / 62 / 3,277 / 29	6.45% (1.79, 20.72)	98.14% (97.63, 98.55)	3.13% (0.86, 10.70)
GLM-4.5	Other structural abnormalities	56	5 / 34 / 3,280 / 51	8.93% (3.87, 19.26)	98.97% (98.57, 99.26)	12.82% (5.60, 26.71)
GLM-4.5	Cardiac position abnormalities	2	0 / 42 / 3,326 / 2	0.00% (0.00, 65.76)	98.75% (98.32, 99.08)	0.00% (0.00, 8.38)
Llama-3.1-70B	Septal defects	139	15 / 42 / 3,189 / 124	10.79% (6.65, 17.04)	98.70% (98.25, 99.04)	26.32% (16.65, 38.98)
Llama-3.1-70B	Great vessels/aortic arch	123	5 / 42 / 3,205 / 118	4.07% (1.75, 9.16)	98.71% (98.26, 99.04)	10.64% (4.63, 22.59)
Llama-3.1-70B	Venous return/systemic veins	34	0 / 44 / 3,292 / 34	0.00% (0.00, 10.15)	98.68% (98.23, 99.02)	0.00% (0.00, 8.03)
Llama-3.1-70B	Valves/outflow tracts	45	2 / 45 / 3,280 / 43	4.44% (1.23, 14.83)	98.65% (98.19, 98.99)	4.26% (1.17, 14.25)
Llama-3.1-70B	Physiological shunt abnormalities	31	2 / 51 / 3,288 / 29	6.45% (1.79, 20.72)	98.47% (98.00, 98.84)	3.77% (1.04, 12.75)
Llama-3.1-70B	Other structural abnormalities	56	4 / 46 / 3,268 / 52	7.14% (2.81, 16.98)	98.61% (98.15, 98.96)	8.00% (3.15, 18.84)
Llama-3.1-70B	Cardiac position abnormalities	2	0 / 62 / 3,306 / 2	0.00% (0.00, 65.76)	98.16% (97.65, 98.56)	0.00% (0.00, 5.83)
Qwen3-Max	Septal defects	139	21 / 41 / 3,190 / 118	15.11% (10.10, 21.99)	98.73% (98.28, 99.06)	33.87% (23.34, 46.28)
Qwen3-Max	Great vessels/aortic arch	123	10 / 43 / 3,204 / 113	8.13% (4.48, 14.32)	98.68% (98.22, 99.02)	18.87% (10.59, 31.36)
Qwen3-Max	Venous return/systemic veins	34	2 / 54 / 3,282 / 32	5.88% (1.63, 19.09)	98.38% (97.89, 98.76)	3.57% (0.98, 12.12)
Qwen3-Max	Valves/outflow tracts	45	4 / 54 / 3,271 / 41	8.89% (3.51, 20.73)	98.38% (97.89, 98.75)	6.90% (2.71, 16.43)
Qwen3-Max	Physiological shunt abnormalities	31	3 / 53 / 3,286 / 28	9.68% (3.35, 24.90)	98.41% (97.93, 98.78)	5.36% (1.84, 14.61)
Qwen3-Max	Other structural abnormalities	56	6 / 41 / 3,273 / 50	10.71% (5.00, 21.47)	98.76% (98.33, 99.09)	12.77% (5.98, 25.17)
Qwen3-Max	Cardiac position abnormalities	2	0 / 40 / 3,328 / 2	0.00% (0.00, 65.76)	98.81% (98.39, 99.13)	0.00% (0.00, 8.76)
GPT-5	Septal defects	139	24 / 34 / 3,197 / 115	17.27% (11.89, 24.41)	98.95% (98.53, 99.25)	41.38% (29.63, 54.20)
GPT-5	Great vessels/aortic arch	123	12 / 35 / 3,212 / 111	9.76% (5.67, 16.28)	98.92% (98.50, 99.22)	25.53% (15.25, 39.51)
GPT-5	Venous return/systemic veins	34	2 / 33 / 3,303 / 32	5.88% (1.63, 19.09)	99.01% (98.61, 99.29)	5.71% (1.58, 18.61)
GPT-5	Valves/outflow tracts	45	5 / 32 / 3,293 / 40	11.11% (4.84, 23.50)	99.04% (98.64, 99.32)	13.51% (5.91, 27.98)

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
GPT-5	Physiological shunt abnormalities	31	4 / 29 / 3,310 / 27	12.90% (5.13, 28.85)	99.13% (98.76, 99.39)	12.12% (4.82, 27.33)
GPT-5	Other structural abnormalities	56	7 / 27 / 3,287 / 49	12.50% (6.19, 23.63)	99.19% (98.82, 99.44)	20.59% (10.35, 36.80)
GPT-5	Cardiac position abnormalities	2	0 / 54 / 3,314 / 2	0.00% (0.00, 65.76)	98.40% (97.91, 98.77)	0.00% (0.00, 6.64)

Multicenter cohort: Full

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
Kimi-K2	Septal defects	139	104 / 5 / 3,226 / 35	74.82% (67.00, 81.30)	99.85% (99.64, 99.93)	95.41% (89.71, 98.02)
Kimi-K2	Great vessels/aortic arch	123	71 / 4 / 3,243 / 52	57.72% (48.89, 66.09)	99.88% (99.68, 99.95)	94.67% (87.07, 97.91)
Kimi-K2	Venous return/systemic veins	34	14 / 7 / 3,329 / 20	41.18% (26.37, 57.78)	99.79% (99.57, 99.90)	66.67% (45.37, 82.81)
Kimi-K2	Valves/outflow tracts	45	27 / 10 / 3,315 / 18	60.00% (45.45, 72.98)	99.70% (99.45, 99.84)	72.97% (57.02, 84.60)
Kimi-K2	Physiological shunt abnormalities	31	16 / 1 / 3,338 / 15	51.61% (34.84, 68.03)	99.97% (99.83, 99.99)	94.12% (73.02, 98.95)
Kimi-K2	Other structural abnormalities	56	31 / 2 / 3,312 / 25	55.36% (42.41, 67.61)	99.94% (99.78, 99.98)	93.94% (80.39, 98.32)
Kimi-K2	Cardiac position abnormalities	2	1 / 3 / 3,365 / 1	50.00% (9.45, 90.55)	99.91% (99.74, 99.97)	25.00% (4.56, 69.94)
DeepSeek-V3.2	Septal defects	139	97 / 6 / 3,225 / 42	69.78% (61.70, 76.80)	99.81% (99.60, 99.91)	94.17% (87.87, 97.30)
DeepSeek-V3.2	Great vessels/aortic arch	123	65 / 3 / 3,244 / 58	52.85% (44.07, 61.45)	99.91% (99.73, 99.97)	95.59% (87.81, 98.49)
DeepSeek-V3.2	Venous return/systemic veins	34	12 / 3 / 3,333 / 22	35.29% (21.49, 52.09)	99.91% (99.74, 99.97)	80.00% (54.81, 92.95)
DeepSeek-V3.2	Valves/outflow tracts	45	25 / 4 / 3,321 / 20	55.56% (41.18, 69.06)	99.88% (99.69, 99.95)	86.21% (69.44, 94.50)
DeepSeek-V3.2	Physiological shunt abnormalities	31	14 / 2 / 3,337 / 17	45.16% (29.16, 62.23)	99.94% (99.78, 99.98)	87.50% (63.98, 96.50)
DeepSeek-V3.2	Other structural abnormalities	56	28 / 6 / 3,308 / 28	50.00% (37.33, 62.67)	99.82% (99.61, 99.92)	82.35% (66.49, 91.65)
DeepSeek-V3.2	Cardiac position abnormalities	2	1 / 2 / 3,366 / 1	50.00% (9.45, 90.55)	99.94% (99.78, 99.98)	33.33% (6.15, 79.23)
GLM-4.5	Septal defects	139	97 / 4 / 3,227 / 42	69.78% (61.70, 76.80)	99.88% (99.68, 99.95)	96.04% (90.26, 98.45)
GLM-4.5	Great vessels/aortic arch	123	65 / 3 / 3,244 / 58	52.85% (44.07, 61.45)	99.91% (99.73, 99.97)	95.59% (87.81, 98.49)
GLM-4.5	Venous return/systemic veins	34	12 / 3 / 3,333 / 22	35.29% (21.49, 52.09)	99.91% (99.74, 99.97)	80.00% (54.81, 92.95)
GLM-4.5	Valves/outflow tracts	45	25 / 7 / 3,318 / 20	55.56% (41.18, 69.06)	99.79% (99.57, 99.90)	78.13% (61.25, 88.98)
GLM-4.5	Physiological shunt abnormalities	31	14 / 7 / 3,332 / 17	45.16% (29.16, 62.23)	99.79% (99.57, 99.90)	66.67% (45.37, 82.81)
GLM-4.5	Other structural abnormalities	56	28 / 1 / 3,313 / 28	50.00% (37.33, 62.67)	99.97% (99.83, 99.99)	96.55% (82.82, 99.39)
GLM-4.5	Cardiac position abnormalities	2	1 / 7 / 3,361 / 1	50.00% (9.45, 90.55)	99.79% (99.57, 99.90)	12.50% (2.24, 47.09)
Llama-3.1-70B	Septal defects	139	95 / 3 / 3,228 / 44	68.35% (60.21, 75.50)	99.91% (99.73, 99.97)	96.94% (91.38, 98.95)
Llama-3.1-70B	Great vessels/aortic arch	123	63 / 11 / 3,236 / 60	51.22% (42.48, 59.88)	99.66% (99.39, 99.81)	85.14% (75.31, 91.49)
Llama-3.1-70B	Venous return/systemic veins	34	11 / 4 / 3,332 / 23	32.35% (19.13, 49.16)	99.88% (99.69, 99.95)	73.33% (48.05, 89.10)
Llama-3.1-70B	Valves/outflow tracts	45	24 / 0 / 3,325 / 21	53.33% (39.08, 67.06)	100.00% (99.88, 100.00)	100.00% (86.20, 100.00)

Model	Disease family	Reference-positive cases	TP / FP / TN / FN	Disease-family recognition performance		
				Recall (95% CI)	Specificity (95% CI)	Precision (95% CI)
Llama-3.1-70B	Physiological shunt abnormalities	31	13 / 6 / 3,333 / 18	41.94% (26.42, 59.23)	99.82% (99.61, 99.92)	68.42% (46.01, 84.64)
Llama-3.1-70B	Other structural abnormalities	56	27 / 5 / 3,309 / 29	48.21% (35.67, 60.99)	99.85% (99.65, 99.94)	84.38% (68.25, 93.14)
Llama-3.1-70B	Cardiac position abnormalities	2	1 / 3 / 3,365 / 1	50.00% (9.45, 90.55)	99.91% (99.74, 99.97)	25.00% (4.56, 69.94)
Qwen3-Max	Septal defects	139	100 / 3 / 3,228 / 39	71.94% (63.96, 78.74)	99.91% (99.73, 99.97)	97.09% (91.78, 99.00)
Qwen3-Max	Great vessels/aortic arch	123	68 / 5 / 3,242 / 55	55.28% (46.47, 63.78)	99.85% (99.64, 99.93)	93.15% (84.95, 97.04)
Qwen3-Max	Venous return/systemic veins	34	13 / 9 / 3,327 / 21	38.24% (23.90, 54.96)	99.73% (99.49, 99.86)	59.09% (38.73, 76.74)
Qwen3-Max	Valves/outflow tracts	45	26 / 3 / 3,322 / 19	57.78% (43.30, 71.03)	99.91% (99.74, 99.97)	89.66% (73.61, 96.42)
Qwen3-Max	Physiological shunt abnormalities	31	15 / 5 / 3,334 / 16	48.39% (31.97, 65.16)	99.85% (99.65, 99.94)	75.00% (53.13, 88.81)
Qwen3-Max	Other structural abnormalities	56	29 / 5 / 3,309 / 27	51.79% (39.01, 64.33)	99.85% (99.65, 99.94)	85.29% (69.87, 93.55)
Qwen3-Max	Cardiac position abnormalities	2	1 / 6 / 3,362 / 1	50.00% (9.45, 90.55)	99.82% (99.61, 99.92)	14.29% (2.57, 51.31)
GPT-5	Septal defects	139	103 / 4 / 3,227 / 36	74.10% (66.24, 80.67)	99.88% (99.68, 99.95)	96.26% (90.78, 98.54)
GPT-5	Great vessels/aortic arch	123	70 / 5 / 3,242 / 53	56.91% (48.08, 65.32)	99.85% (99.64, 99.93)	93.33% (85.32, 97.12)
GPT-5	Venous return/systemic veins	34	13 / 2 / 3,334 / 21	38.24% (23.90, 54.96)	99.94% (99.78, 99.98)	86.67% (62.12, 96.26)
GPT-5	Valves/outflow tracts	45	27 / 9 / 3,316 / 18	60.00% (45.45, 72.98)	99.73% (99.49, 99.86)	75.00% (58.93, 86.25)
GPT-5	Physiological shunt abnormalities	31	15 / 2 / 3,337 / 16	48.39% (31.97, 65.16)	99.94% (99.78, 99.98)	88.24% (65.66, 96.71)
GPT-5	Other structural abnormalities	56	30 / 3 / 3,311 / 26	53.57% (40.70, 65.98)	99.91% (99.73, 99.97)	90.91% (76.43, 96.86)
GPT-5	Cardiac position abnormalities	2	1 / 4 / 3,364 / 1	50.00% (9.45, 90.55)	99.88% (99.70, 99.95)	20.00% (3.62, 62.45)

Supplementary Table S4 | Disease-family set prediction and stratification by single-family and multiple-family cases

Disease-family sets comprise labels from the seven-family taxonomy. Counts are shown as reference / predicted / correct labels. Micro-averaged metrics use label counts; Jaccard similarity and exact-match rate are summarized at the case level.

Prospective cohort: all positive cases (n = 340)

Model	Condition	Reference / predicted / correct labels	Family-set metrics			Case-level agreement	
			Micro-averaged recall (95% CI)	Micro-averaged precision (95% CI)	Micro-averaged F1 (95% CI)	Mean Jaccard (95% CI)	Exact-match rate (95% CI)

Model	Condition	Reference / predicted / correct labels	Family-set metrics			Case-level agreement	
			Micro-averaged recall (95% CI)	Micro-averaged precision (95% CI)	Micro-averaged F1 (95% CI)	Mean Jaccard (95% CI)	Exact-match rate (95% CI)
Kimi-K2	Base	495 / 122 / 81	16.36% (12.71, 20.17)	66.39% (57.25, 75.46)	26.26% (21.02, 31.33)	16.99% (13.21, 20.76)	13.53% (10.30, 17.58)
Kimi-K2	Full	**495 / 349 / 312**	**63.03% (57.77, 68.04)**	**89.40% (86.31, 92.38)**	**73.93% (69.86, 77.78)**	**61.86% (57.17, 66.54)**	**52.65% (47.34, 57.90)**
DeepSeek-V3.2	Base	495 / 119 / 55	11.11% (7.98, 14.38)	46.22% (36.89, 56.20)	17.92% (12.96, 22.41)	10.39% (7.38, 13.40)	7.65% (5.27, 10.97)
DeepSeek-V3.2	Full	**495 / 321 / 285**	**57.58% (52.59, 62.53)**	**88.79% (85.46, 91.82)**	**69.85% (65.79, 73.74)**	**55.26% (50.41, 60.11)**	**46.76% (41.53, 52.07)**
GLM-4.5	Base	495 / 100 / 56	11.31% (8.42, 14.46)	56.00% (45.95, 65.60)	18.82% (14.67, 23.41)	11.76% (8.61, 14.92)	8.53% (6.00, 11.98)
GLM-4.5	Full	**495 / 326 / 287**	**57.98% (52.82, 62.93)**	**88.04% (84.42, 91.27)**	**69.91% (65.67, 74.02)**	**56.43% (51.58, 61.28)**	**48.24% (42.97, 53.54)**
Llama-3.1-70B	Base	495 / 93 / 46	9.29% (6.34, 12.57)	49.46% (38.68, 60.42)	15.65% (10.97, 20.75)	8.09% (5.42, 10.75)	5.59% (3.61, 8.56)
Llama-3.1-70B	Full	**495 / 308 / 278**	**56.16% (51.10, 61.23)**	**90.26% (87.00, 93.31)**	**69.24% (65.01, 73.26)**	**54.04% (49.13, 58.95)**	**46.47% (41.24, 51.78)**
Qwen3-Max	Base	495 / 100 / 66	13.33% (10.12, 16.70)	66.00% (57.43, 75.00)	22.18% (17.35, 27.05)	12.67% (9.41, 15.93)	9.41% (6.75, 12.98)
Qwen3-Max	Full	**495 / 323 / 297**	**60.00% (55.23, 64.99)**	**91.95% (89.03, 94.83)**	**72.62% (68.53, 76.38)**	**59.17% (54.35, 63.99)**	**51.18% (45.88, 56.45)**
GPT-5	Base	495 / 135 / 76	15.35% (11.65, 19.22)	56.30% (48.00, 64.44)	24.13% (19.19, 29.04)	13.74% (10.38, 17.09)	9.71% (6.99, 13.32)
GPT-5	Full	**495 / 345 / 308**	**62.22% (57.00, 67.20)**	**89.28% (86.02, 92.55)**	**73.33% (69.25, 77.32)**	**60.48% (55.72, 65.24)**	**51.76% (46.46, 57.03)**

Prospective cohort: single-family cases (n = 226)

Model	Condition	Reference / predicted / correct labels	Family-set metrics			Case-level agreement	
			Micro-averaged recall (95% CI)	Micro-averaged precision (95% CI)	Micro-averaged F1 (95% CI)	Mean Jaccard (95% CI)	Exact-match rate (95% CI)
Kimi-K2	Base	226 / 77 / 45	19.91% (15.04, 25.22)	58.44% (46.91, 70.42)	29.70% (23.05, 36.61)	19.03% (13.94, 24.11)	18.14% (13.66, 23.68)
Kimi-K2	Full	**226 / 184 / 154**	**68.14% (61.95, 74.34)**	**83.70% (78.77, 88.42)**	**75.12% (70.53, 79.34)**	**63.20% (57.20, 69.20)**	**58.41% (51.89, 64.64)**
DeepSeek-V3.2	Base	226 / 74 / 27	11.95% (7.96, 16.81)	36.49% (25.63, 47.31)	18.00% (12.20, 24.16)	10.62% (6.72, 14.52)	9.29% (6.16, 13.79)
DeepSeek-V3.2	Full	**226 / 164 / 136**	**60.18% (53.98, 66.37)**	**82.93% (78.02, 88.07)**	**69.74% (64.17, 74.50)**	**55.97% (49.74, 62.21)**	**51.77% (45.28, 58.20)**
GLM-4.5	Base	226 / 70 / 34	15.04% (10.62, 19.91)	48.57% (37.10, 59.65)	22.97% (16.44, 29.13)	13.72% (9.33, 18.11)	12.39% (8.71, 17.32)
GLM-4.5	Full	**226 / 170 / 141**	**62.39% (55.75, 68.58)**	**82.94% (77.83, 88.31)**	**71.21% (66.15, 75.57)**	**58.04% (51.84, 64.24)**	**53.98% (47.47, 60.36)**
Llama-3.1-70B	Base	226 / 56 / 20	8.85% (5.31, 12.83)	35.71% (24.53, 47.92)	14.18% (8.69, 19.86)	7.52% (4.23, 10.82)	6.19% (3.73, 10.13)
Llama-3.1-70B	Full	**226 / 147 / 128**	**56.64% (50.44, 63.27)**	**87.07% (82.12, 91.94)**	**68.63% (63.30, 73.77)**	**53.69% (47.33, 60.05)**	**50.88% (44.41, 57.33)**
Qwen3-Max	Base	226 / 55 / 32	14.16% (10.18, 19.03)	58.18% (45.83, 71.11)	22.78% (16.42, 29.04)	12.98% (8.67, 17.29)	11.95% (8.34, 16.82)
Qwen3-Max	Full	**226 / 161 / 142**	**62.83% (56.19, 68.58)**	**88.20% (83.44, 93.10)**	**73.39% (68.28, 78.09)**	**60.32% (54.06, 66.59)**	**57.96% (51.45, 64.21)**
GPT-5	Base	226 / 76 / 35	15.49% (10.62, 19.91)	46.05% (35.94, 56.72)	23.18% (17.18, 29.56)	13.42% (9.14, 17.71)	11.50% (7.97, 16.32)
GPT-5	Full	**226 / 180 / 148**	**65.49% (58.85, 71.68)**	**82.22% (76.76, 87.28)**	**72.91% (67.84, 77.70)**	**61.14% (55.02, 67.25)**	**57.08% (50.56, 63.36)**

Prospective cohort: multiple-family cases (n = 114)

Model	Condition	Reference / predicted / correct labels	Family-set metrics			Case-level agreement	
			Micro-averaged recall (95% CI)	Micro-averaged precision (95% CI)	Micro-averaged F1 (95% CI)	Mean Jaccard (95% CI)	Exact-match rate (95% CI)
Kimi-K2	Base	269 / 45 / 36	13.38% (8.61, 18.66)	80.00% (66.67, 91.67)	22.93% (14.91, 30.63)	12.94% (7.90, 17.98)	4.39% (1.89, 9.86)
Kimi-K2	Full	**269 / 165 / 158**	**58.74% (51.26, 66.54)**	**95.76% (92.74, 98.24)**	**72.81% (66.34, 78.59)**	**59.20% (51.76, 66.63)**	**41.23% (32.62, 50.41)**
DeepSeek-V3.2	Base	269 / 45 / 28	10.41% (5.86, 15.29)	62.22% (45.45, 76.60)	17.83% (10.74, 25.30)	9.94% (5.31, 14.57)	4.39% (1.89, 9.86)
DeepSeek-V3.2	Full	**269 / 157 / 149**	**55.39% (48.13, 63.20)**	**94.90% (91.16, 98.03)**	**69.95% (63.03, 75.82)**	**53.85% (46.20, 61.49)**	**36.84% (28.55, 45.99)**
GLM-4.5	Base	269 / 30 / 22	8.18% (4.60, 12.36)	73.33% (57.13, 88.00)	14.72% (8.45, 20.67)	7.89% (4.33, 11.46)	0.88% (0.16, 4.80)
GLM-4.5	Full	**269 / 156 / 146**	**54.28% (46.07, 62.50)**	**93.59% (89.93, 96.97)**	**68.71% (62.07, 74.83)**	**53.23% (45.51, 60.95)**	**36.84% (28.55, 45.99)**
Llama-3.1-70B	Base	269 / 37 / 26	9.67% (5.06, 14.70)	70.27% (51.52, 85.19)	16.99% (9.46, 24.84)	9.21% (4.63, 13.79)	4.39% (1.89, 9.86)
Llama-3.1-70B	Full	**269 / 161 / 150**	**55.76% (48.89, 63.43)**	**93.17% (89.29, 96.60)**	**69.77% (63.37, 75.57)**	**54.74% (47.17, 62.30)**	**37.72% (29.36, 46.88)**
Qwen3-Max	Base	269 / 45 / 34	12.64% (8.11, 17.99)	75.56% (65.00, 85.71)	21.66% (14.71, 28.93)	12.06% (7.32, 16.80)	4.39% (1.89, 9.86)
Qwen3-Max	Full	**269 / 162 / 155**	**57.62% (50.38, 65.11)**	**95.68% (92.55, 98.61)**	**71.93% (65.75, 77.53)**	**56.89% (49.53, 64.24)**	**37.72% (29.36, 46.88)**
GPT-5	Base	269 / 59 / 41	15.24% (10.04, 21.13)	69.49% (56.71, 81.16)	25.00% (17.03, 32.95)	14.36% (8.98, 19.73)	6.14% (3.01, 12.13)
GPT-5	Full	**269 / 165 / 160**	**59.48% (51.68, 66.92)**	**96.97% (94.22, 99.35)**	**73.73% (67.61, 79.30)**	**59.18% (51.66, 66.70)**	**41.23% (32.62, 50.41)**

Multicenter cohort: all positive cases (n = 316)

Model	Condition	Reference / predicted / correct labels	Family-set metrics			Case-level agreement	
			Micro-averaged recall (95% CI)	Micro-averaged precision (95% CI)	Micro-averaged F1 (95% CI)	Mean Jaccard (95% CI)	Exact-match rate (95% CI)
Kimi-K2	Base	430 / 102 / 59	13.72% (10.30, 17.65)	57.84% (49.06, 67.06)	22.18% (17.18, 27.51)	12.24% (8.91, 15.56)	8.86% (6.20, 12.51)
Kimi-K2	Full	**430 / 291 / 264**	**61.40% (56.63, 66.14)**	**90.72% (86.97, 94.16)**	**73.23% (69.28, 77.05)**	**57.35% (52.36, 62.34)**	**48.73% (43.27, 54.23)**
DeepSeek-V3.2	Base	430 / 99 / 35	8.14% (5.58, 10.98)	35.35% (26.09, 44.44)	13.23% (9.27, 17.25)	7.50% (4.87, 10.13)	5.06% (3.14, 8.07)
DeepSeek-V3.2	Full	**430 / 260 / 242**	**56.28% (50.95, 61.57)**	**93.08% (90.00, 95.89)**	**70.14% (65.57, 74.32)**	**55.46% (50.24, 60.68)**	**50.32% (44.83, 55.79)**
GLM-4.5	Base	430 / 83 / 36	8.37% (5.65, 11.06)	43.37% (32.14, 53.69)	14.04% (9.73, 18.61)	7.20% (4.66, 9.74)	4.43% (2.66, 7.30)
GLM-4.5	Full	**430 / 266 / 242**	**56.28% (51.00, 61.48)**	**90.98% (87.46, 94.09)**	**69.54% (65.01, 73.86)**	**54.85% (49.75, 59.95)**	**47.47% (42.03, 52.97)**
Llama-3.1-70B	Base	430 / 86 / 28	6.51% (4.19, 9.13)	32.56% (23.46, 42.47)	10.85% (7.13, 14.77)	6.49% (3.99, 8.98)	4.43% (2.66, 7.30)
Llama-3.1-70B	Full	**430 / 258 / 234**	**54.42% (49.18, 59.45)**	**90.70% (87.22, 93.95)**	**68.02% (63.78, 72.17)**	**51.56% (46.49, 56.62)**	**43.04% (37.69, 48.55)**
Qwen3-Max	Base	430 / 95 / 46	10.70% (7.73, 13.84)	48.42% (38.98, 57.66)	17.52% (13.00, 22.26)	8.97% (6.19, 11.74)	5.38% (3.39, 8.45)
Qwen3-Max	Full	**430 / 280 / 252**	**58.60% (53.50, 63.60)**	**90.00% (86.32, 93.41)**	**70.99% (66.96, 74.93)**	**56.28% (51.31, 61.24)**	**46.84% (41.40, 52.34)**
GPT-5	Base	430 / 106 / 54	12.56% (9.29, 15.93)	50.94% (41.30, 61.26)	20.15% (15.21, 24.82)	12.18% (8.83, 15.53)	9.18% (6.47, 12.87)
GPT-5	Full	**430 / 284 / 259**	**60.23% (54.94, 65.54)**	**91.20% (87.84, 94.29)**	**72.55% (68.29, 76.71)**	**60.34% (55.30, 65.37)**	**53.80% (48.29, 59.22)**

Multicenter cohort: single-family cases (n = 229)

Model	Condition	Reference / predicted / correct labels	Family-set metrics			Case-level agreement	
			Micro-averaged recall (95% CI)	Micro-averaged precision (95% CI)	Micro-averaged F1 (95% CI)	Mean Jaccard (95% CI)	Exact-match rate (95% CI)
Kimi-K2	Base	229 / 61 / 31	13.54% (9.17, 18.34)	50.82% (40.00, 61.91)	21.38% (15.38, 27.52)	11.72% (7.70, 15.74)	10.04% (6.79, 14.62)
Kimi-K2	Full	**229 / 155 / 134**	**58.52% (51.97, 64.63)**	**86.45% (81.32, 91.55)**	**69.79% (64.36, 74.56)**	**54.95% (48.69, 61.21)**	**51.53% (45.08, 57.92)**
DeepSeek-V3.2	Base	229 / 72 / 24	10.48% (6.99, 14.85)	33.33% (24.05, 42.86)	15.95% (10.42, 21.29)	8.22% (4.91, 11.54)	6.11% (3.68, 10.00)
DeepSeek-V3.2	Full	**229 / 143 / 131**	**57.21% (50.66, 63.76)**	**91.61% (87.14, 95.89)**	**70.43% (64.95, 75.59)**	**56.33% (49.92, 62.75)**	**55.46% (48.98, 61.75)**
GLM-4.5	Base	229 / 57 / 18	7.86% (4.37, 11.79)	31.58% (20.37, 42.86)	12.59% (7.55, 17.94)	6.48% (3.44, 9.51)	5.24% (3.02, 8.93)
GLM-4.5	Full	**229 / 148 / 130**	**56.77% (50.66, 62.89)**	**87.84% (82.72, 92.91)**	**68.97% (63.24, 74.07)**	**54.59% (48.23, 60.94)**	**52.40% (45.95, 58.78)**
Llama-3.1-70B	Base	229 / 66 / 23	10.04% (6.11, 13.97)	34.85% (24.07, 45.31)	15.59% (10.18, 21.33)	8.08% (4.77, 11.39)	6.11% (3.68, 10.00)
Llama-3.1-70B	Full	**229 / 142 / 123**	**53.71% (47.60, 59.83)**	**86.62% (81.58, 91.67)**	**66.31% (60.87, 71.36)**	**50.44% (44.13, 56.74)**	**47.16% (40.80, 53.62)**
Qwen3-Max	Base	229 / 64 / 25	10.92% (6.99, 14.85)	39.06% (29.03, 50.00)	17.06% (11.68, 22.44)	8.52% (5.13, 11.90)	6.55% (4.01, 10.52)
Qwen3-Max	Full	**229 / 160 / 136**	**59.39% (52.84, 65.50)**	**85.00% (79.65, 90.28)**	**69.92% (64.80, 74.62)**	**55.75% (49.50, 62.00)**	**52.40% (45.95, 58.78)**
GPT-5	Base	229 / 78 / 33	14.41% (10.04, 18.78)	42.31% (32.31, 53.25)	21.50% (15.59, 27.54)	13.03% (8.76, 17.29)	11.79% (8.23, 16.61)
GPT-5	Full	**229 / 172 / 151**	**65.94% (59.39, 72.05)**	**87.79% (83.15, 92.17)**	**75.31% (70.74, 80.00)**	**62.88% (56.79, 68.98)**	**59.83% (53.36, 65.96)**

Multicenter cohort: multiple-family cases (n = 87)

Model	Condition	Reference / predicted / correct labels	Family-set metrics			Case-level agreement	
			Micro-averaged recall (95% CI)	Micro-averaged precision (95% CI)	Micro-averaged F1 (95% CI)	Mean Jaccard (95% CI)	Exact-match rate (95% CI)
Kimi-K2	Base	201 / 41 / 28	13.93% (8.33, 20.31)	68.29% (54.28, 81.08)	23.14% (14.85, 31.37)	13.60% (7.65, 19.55)	5.75% (2.48, 12.76)
Kimi-K2	Full	**201 / 136 / 130**	**64.68% (56.59, 72.28)**	**95.59% (90.67, 99.27)**	**77.15% (71.07, 82.64)**	**63.68% (56.14, 71.22)**	**41.38% (31.61, 51.88)**
DeepSeek-V3.2	Base	201 / 27 / 11	5.47% (1.98, 9.41)	40.74% (19.35, 63.16)	9.65% (4.33, 16.10)	5.59% (1.62, 9.57)	2.30% (0.63, 8.00)
DeepSeek-V3.2	Full	**201 / 117 / 111**	**55.22% (46.12, 63.68)**	**94.87% (90.91, 98.33)**	**69.81% (61.86, 76.92)**	**53.16% (44.38, 61.95)**	**36.78% (27.41, 47.27)**
GLM-4.5	Base	201 / 26 / 18	8.96% (4.90, 13.30)	69.23% (48.00, 88.89)	15.86% (9.13, 22.81)	9.10% (4.43, 13.76)	2.30% (0.63, 8.00)
GLM-4.5	Full	**201 / 118 / 112**	**55.72% (47.42, 64.22)**	**94.92% (91.06, 98.35)**	**70.22% (63.22, 76.70)**	**55.56% (47.38, 63.73)**	**34.48% (25.34, 44.94)**
Llama-3.1-70B	Base	201 / 20 / 5	2.49% (0.49, 5.39)	25.00% (5.00, 50.00)	4.52% (0.91, 9.13)	2.30% (0.00, 4.61)	0.00% (0.00, 4.23)
Llama-3.1-70B	Full	**201 / 116 / 111**	**55.22% (46.73, 63.16)**	**95.69% (91.85, 99.10)**	**70.03% (62.54, 76.69)**	**54.50% (46.44, 62.56)**	**32.18% (23.30, 42.57)**
Qwen3-Max	Base	201 / 31 / 21	10.45% (6.03, 15.76)	67.74% (50.00, 83.87)	18.10% (10.57, 25.94)	10.15% (5.36, 14.95)	2.30% (0.63, 8.00)
Qwen3-Max	Full	**201 / 120 / 116**	**57.71% (49.52, 65.40)**	**96.67% (92.80, 99.21)**	**72.27% (65.33, 78.31)**	**57.66% (50.08, 65.24)**	**32.18% (23.30, 42.57)**
GPT-5	Base	201 / 28 / 21	10.45% (6.22, 15.61)	75.00% (58.82, 89.47)	18.34% (11.06, 25.60)	9.96% (5.17, 14.75)	2.30% (0.63, 8.00)
GPT-5	Full	**201 / 112 / 108**	**53.73% (45.45, 62.62)**	**96.43% (92.38, 99.17)**	**69.01% (60.77, 76.45)**	**53.64% (44.83, 62.45)**	**37.93% (28.45, 48.43)**

Supplementary Table S5 | Paired case-level comparisons of direct base-model generation and the full system

a denotes cases classified correctly under both conditions; b, cases classified correctly by Base but incorrectly by Full; c, cases classified incorrectly by Base but correctly by Full; and d, cases classified incorrectly under both conditions. Differences are Full minus Base, using the appropriate denominator for sensitivity, specificity or accuracy. The 95% confidence interval for each paired difference combines the bounds of the separate 97.5% Clopper–Pearson intervals for the two discordant-pair proportions. P (exact) denotes the two-sided exact McNemar test; P (Holm) denotes Holm adjustment across 18 paired tests within each cohort.

Prospective cohort

Model	Outcome	n	Paired case counts				Effect estimate	Significance tests	
			a	b	c	d	Paired difference (95% CI)	P (exact)	P (Holm)
Kimi-K2	Sensitivity	340	181	23	131	5	0.318 (0.222, 0.406)	1.49E-19	7.47E-19
Kimi-K2	Specificity	1,118	866	15	235	2	0.197 (0.160, 0.232)	5.48E-52	4.38E-51
Kimi-K2	Accuracy	1,458	1,047	38	366	7	0.225 (0.189, 0.260)	1.89E-68	2.08E-67
DeepSeek-V3.2	Sensitivity	340	237	12	74	17	0.182 (0.105, 0.256)	4.69E-12	4.69E-12
DeepSeek-V3.2	Specificity	1,118	263	13	840	2	0.740 (0.700, 0.774)	6.27E-229	1.07E-227
DeepSeek-V3.2	Accuracy	1,458	500	25	914	19	0.610 (0.571, 0.645)	4.28E-234	7.71E-233
GLM-4.5	Sensitivity	340	200	21	108	11	0.256 (0.164, 0.342)	2.67E-15	5.33E-15
GLM-4.5	Specificity	1,118	791	14	307	6	0.262 (0.223, 0.299)	5.20E-73	6.24E-72
GLM-4.5	Accuracy	1,458	991	35	415	17	0.261 (0.224, 0.296)	1.36E-83	1.91E-82
Llama-3.1-70B	Sensitivity	340	187	17	115	21	0.288 (0.198, 0.372)	4.62E-19	1.85E-18
Llama-3.1-70B	Specificity	1,118	791	12	305	10	0.262 (0.223, 0.299)	1.36E-74	1.76E-73
Llama-3.1-70B	Accuracy	1,458	978	29	420	31	0.268 (0.232, 0.303)	5.45E-90	8.18E-89
Qwen3-Max	Sensitivity	340	163	18	151	8	0.391 (0.296, 0.477)	2.34E-27	1.40E-26
Qwen3-Max	Specificity	1,118	832	10	274	2	0.236 (0.199, 0.271)	5.35E-68	5.35E-67
Qwen3-Max	Accuracy	1,458	995	28	425	10	0.272 (0.236, 0.307)	3.02E-92	4.84E-91
GPT-5	Sensitivity	340	225	13	91	11	0.229 (0.147, 0.307)	1.40E-15	4.20E-15
GPT-5	Specificity	1,118	882	12	223	1	0.189 (0.153, 0.223)	1.70E-51	1.19E-50
GPT-5	Accuracy	1,458	1,107	25	314	12	0.198 (0.165, 0.230)	9.08E-65	8.17E-64

Multicenter cohort

Model	Outcome	n	Paired case counts				Effect estimate	Significance tests	
			a	b	c	d	Paired difference (95% CI)	P (exact)	P (Holm)
Kimi-K2	Sensitivity	316	168	12	113	23	0.320 (0.228, 0.403)	9.25E-22	4.62E-21
Kimi-K2	Specificity	3,054	2,360	22	624	48	0.197 (0.177, 0.217)	2.94E-154	2.35E-153
Kimi-K2	Accuracy	3,370	2,528	34	737	71	0.209 (0.188, 0.228)	3.95E-173	3.95E-172
DeepSeek-V3.2	Sensitivity	316	203	19	77	17	0.184 (0.094, 0.268)	1.86E-09	1.86E-09

Model	Outcome	n	Paired case counts				Effect estimate	Significance tests	
			a	b	c	d	Paired difference (95% CI)	P (exact)	P (Holm)
DeepSeek-V3.2	Specificity	3,054	711	19	2,279	45	0.740 (0.718, 0.760)	<1e-300	<1e-300
DeepSeek-V3.2	Accuracy	3,370	914	38	2,356	62	0.688 (0.665, 0.709)	<1e-300	<1e-300
GLM-4.5	Sensitivity	316	184	12	93	27	0.256 (0.168, 0.338)	1.10E-16	3.30E-16
GLM-4.5	Specificity	3,054	2,154	20	821	59	0.262 (0.240, 0.284)	1.43E-213	1.72E-212
GLM-4.5	Accuracy	3,370	2,338	32	914	86	0.262 (0.240, 0.283)	1.32E-225	1.98E-224
Llama-3.1-70B	Sensitivity	316	165	15	106	30	0.288 (0.195, 0.374)	4.72E-18	1.89E-17
Llama-3.1-70B	Specificity	3,054	2,156	12	812	74	0.262 (0.241, 0.282)	3.42E-222	4.79E-221
Llama-3.1-70B	Accuracy	3,370	2,321	27	918	104	0.264 (0.243, 0.285)	9.49E-233	1.52E-231
Qwen3-Max	Sensitivity	316	146	13	137	20	0.392 (0.297, 0.478)	2.83E-27	1.70E-26
Qwen3-Max	Specificity	3,054	2,261	14	735	44	0.236 (0.215, 0.256)	1.22E-196	1.34E-195
Qwen3-Max	Accuracy	3,370	2,407	27	872	64	0.251 (0.230, 0.271)	1.71E-219	2.22E-218
GPT-5	Sensitivity	316	199	13	85	19	0.228 (0.141, 0.309)	3.98E-14	7.97E-14
GPT-5	Specificity	3,054	2,400	19	593	42	0.188 (0.168, 0.207)	6.69E-149	4.68E-148
GPT-5	Accuracy	3,370	2,599	32	678	61	0.192 (0.172, 0.211)	1.27E-158	1.14E-157

Supplementary Table S6 | Paired case-level comparisons of the full Qwen3-Max system with comparators and ablation conditions

a denotes cases classified correctly under both conditions; b, cases classified correctly by the comparator but incorrectly by Full; c, cases classified incorrectly by the comparator but correctly by Full; and d, cases classified incorrectly under both conditions. Differences are Full Qwen3-Max minus the comparator. Paired-difference intervals and tests follow the methods in Supplementary Table S5. P (Holm) is adjusted across 12 comparisons within each cohort.

Prospective cohort

Comparator	n	Paired case counts				Effect estimate	Significance tests	
		a	b	c	d	Paired difference (95% CI)	P (exact)	P (Holm)

Comparator	n	Paired case counts				Effect estimate	Significance tests	
		a	b	c	d	Paired difference (95% CI)	P (exact)	P (Holm)
GA rules	1,458	1,319	4	101	34	0.067 (0.047, 0.085)	2.45E-25	2.45E-24
kNN	1,458	1,348	6	72	32	0.045 (0.028, 0.062)	1.85E-15	1.48E-14
Logistic regression	1,458	1,321	5	99	33	0.064 (0.045, 0.083)	9.54E-24	8.59E-23
Text RAG	1,458	1,224	2	196	36	0.133 (0.109, 0.156)	9.81E-56	1.08E-54
Structured RAG no validation	1,458	1,360	13	60	25	0.032 (0.014, 0.050)	2.31E-08	1.12E-07
No GA normalization	1,458	1,371	9	49	29	0.027 (0.011, 0.043)	8.96E-08	2.69E-07
No polarity control	1,458	1,360	10	60	28	0.034 (0.017, 0.051)	8.00E-10	4.80E-09
No SAWS threshold	1,458	1,378	15	42	23	0.019 (0.002, 0.035)	4.60E-04	9.20E-04
No closure	1,458	1,362	12	58	26	0.032 (0.014, 0.049)	2.25E-08	1.12E-07
kNN + template	1,458	1,348	6	72	32	0.045 (0.028, 0.062)	1.85E-15	1.48E-14
Matched evidence + template	1,458	1,420	0	0	38	0.000 (-0.003, 0.003)	1.00E+00	1.00E+00
Direct Qwen3-Max	1,458	995	28	425	10	0.272 (0.236, 0.307)	3.02E-92	3.63E-91

Multicenter cohort

Comparator	n	Paired case counts				Effect estimate	Significance tests	
		a	b	c	d	Paired difference (95% CI)	P (exact)	P (Holm)
GA rules	3,370	3,076	13	203	78	0.056 (0.044, 0.068)	5.02E-45	5.02E-44
kNN	3,370	3,129	19	150	72	0.039 (0.028, 0.050)	1.88E-26	1.31E-25
Logistic regression	3,370	3,086	16	193	75	0.053 (0.040, 0.064)	9.31E-40	8.38E-39
Text RAG	3,370	2,837	2	442	89	0.131 (0.116, 0.145)	4.35E-129	4.78E-128
Structured RAG no validation	3,370	3,126	20	153	71	0.039 (0.028, 0.051)	1.45E-26	1.16E-25
No GA normalization	3,370	3,179	26	100	65	0.022 (0.012, 0.032)	1.98E-11	3.97E-11
No polarity control	3,370	3,121	29	158	62	0.038 (0.026, 0.050)	1.09E-22	4.36E-22
No SAWS threshold	3,370	3,161	28	118	63	0.027 (0.016, 0.038)	2.40E-14	7.21E-14
No closure	3,370	3,115	26	164	65	0.041 (0.029, 0.053)	1.10E-25	5.52E-25
kNN + template	3,370	3,129	19	150	72	0.039 (0.028, 0.050)	1.88E-26	1.31E-25
Matched evidence + template	3,370	3,279	0	0	91	0.000 (-0.001, 0.001)	1.00E+00	1.00E+00
Direct Qwen3-Max	3,370	2,407	27	872	64	0.251 (0.230, 0.271)	1.71E-219	2.05E-218

Supplementary Table S7 | Calibration of the shared retrieval risk layer

The shared retrieval risk layer is evaluated separately using raw support scores and fixed-mapping probabilities. CITL denotes calibration-in-the-large; intercept and slope denote the calibration intercept and slope.

Prospective cohort

Metric	n	Shared retrieval risk layer	
		Raw support score (95% CI)	Fixed-mapping probability (95% CI)
Brier	1,458	0.086 (0.078, 0.094)	0.067 (0.058, 0.076)
ECE	1,458	0.111 (0.099, 0.125)	0.010 (0.008, 0.027)
CITL	1,458	-0.992 (-1.156, -0.827)	0.027 (-0.176, 0.230)
intercept	1,458	-1.146 (-1.355, -0.938)	0.006 (-0.207, 0.218)
slope	1,458	1.549 (1.374, 1.724)	0.968 (0.859, 1.077)

Multicenter cohort

Metric	n	Shared retrieval risk layer	
		Raw support score (95% CI)	Fixed-mapping probability (95% CI)
Brier	3,370	0.091 (0.085, 0.096)	0.043 (0.038, 0.048)
ECE	3,370	0.174 (0.166, 0.182)	0.005 (0.004, 0.014)
CITL	3,370	-1.918 (-2.059, -1.776)	0.061 (-0.108, 0.231)
intercept	3,370	-2.117 (-2.296, -1.939)	-0.070 (-0.270, 0.130)
slope	3,370	1.444 (1.310, 1.579)	0.903 (0.819, 0.987)

Supplementary Table S8 | Decile-based reliability analysis of the shared risk layer

Each bin reports the number of cases, positive cases, mean predicted probability and observed positive-case proportion. Observed proportions are accompanied by 95% confidence intervals.

Prospective cohort: raw support scores

Bin	n	Positive cases	Mean predicted probability	Observed positive-case proportion (95% CI)
1	146	0	0.015	0.000 (0.000, 0.026)
2	146	0	0.039	0.000 (0.000, 0.026)
3	146	1	0.070	0.007 (0.001, 0.038)
4	145	1	0.115	0.007 (0.001, 0.038)
5	146	7	0.178	0.048 (0.023, 0.096)
6	146	9	0.265	0.062 (0.033, 0.113)
7	145	16	0.399	0.110 (0.069, 0.172)
8	146	55	0.581	0.377 (0.302, 0.458)
9	146	111	0.812	0.760 (0.685, 0.822)
10	146	140	0.950	0.959 (0.913, 0.981)

Prospective cohort: fixed-mapping probabilities

Bin	n	Positive cases	Mean predicted probability	Observed positive-case proportion (95% CI)
1	146	0	0.000	0.000 (0.000, 0.026)
2	146	0	0.002	0.000 (0.000, 0.026)
3	146	1	0.005	0.007 (0.001, 0.038)
4	145	1	0.012	0.007 (0.001, 0.038)
5	146	7	0.026	0.048 (0.023, 0.096)
6	146	9	0.057	0.062 (0.033, 0.113)
7	145	16	0.139	0.110 (0.069, 0.172)
8	146	55	0.349	0.377 (0.302, 0.458)
9	146	111	0.756	0.760 (0.685, 0.822)
10	146	140	0.967	0.959 (0.913, 0.981)

Multicenter cohort: raw support scores

Bin	n	Positive cases	Mean predicted probability	Observed positive-case proportion (95% CI)
1	337	0	0.015	0.000 (0.000, 0.011)
2	337	0	0.038	0.000 (0.000, 0.011)
3	337	2	0.067	0.006 (0.002, 0.021)
4	337	1	0.102	0.003 (0.001, 0.017)
5	337	4	0.147	0.012 (0.005, 0.030)
6	337	5	0.201	0.015 (0.006, 0.034)
7	337	13	0.279	0.039 (0.023, 0.065)
8	337	16	0.394	0.047 (0.029, 0.076)
9	337	54	0.574	0.160 (0.125, 0.203)
10	337	221	0.860	0.656 (0.604, 0.705)

Multicenter cohort: fixed-mapping probabilities

Bin	n	Positive cases	Mean predicted probability	Observed positive-case proportion (95% CI)
1	337	0	0.000	0.000 (0.000, 0.011)
2	337	0	0.001	0.000 (0.000, 0.011)
3	337	2	0.002	0.006 (0.002, 0.021)
4	337	1	0.003	0.003 (0.001, 0.017)
5	337	4	0.006	0.012 (0.005, 0.030)
6	337	5	0.011	0.015 (0.006, 0.034)
7	337	13	0.022	0.039 (0.023, 0.065)
8	337	16	0.051	0.047 (0.029, 0.076)
9	337	54	0.152	0.160 (0.125, 0.203)
10	337	221	0.665	0.656 (0.604, 0.705)

Supplementary Table S9 | Clinical workflow and escalation events for the full Qwen3-Max system

Event rates use all cases in the corresponding cohort as the denominator. Acceptance unchanged, acceptance after minor revision, and major revision or rejection are the three report-handling categories. "Accepted unchanged or after minor revision" is the sum of the first two categories. Results refer to the full ExpEvaFCHD system using Qwen3-Max as the generation backend.

Prospective cohort

Event	Event count	n	Event rate (95% CI)
Accepted unchanged	729	1,458	0.500 (0.474, 0.526)
Accepted after minor revision	396	1,458	0.272 (0.249, 0.295)
Major revision or rejection	333	1,458	0.228 (0.208, 0.251)
Accepted unchanged or after minor revision	**1,125**	1,458	**0.772 (0.749, 0.792)**
Low-evidence alert triggered	66	1,458	0.045 (0.036, 0.057)
Draft-validation failure	86	1,458	0.059 (0.048, 0.072)
Fallback after generation or formatting failure	5	1,458	0.003 (0.001, 0.008)
Physician review or escalation	449	1,458	0.308 (0.285, 0.332)
False negatives without escalation flags	24	1,458	0.016 (0.011, 0.024)

Multicenter cohort

Event	Event count	n	Event rate (95% CI)
Accepted unchanged	1,758	3,370	0.522 (0.505, 0.538)
Accepted after minor revision	887	3,370	0.263 (0.249, 0.278)
Major revision or rejection	725	3,370	0.215 (0.202, 0.229)
Accepted unchanged or after minor revision	**2,645**	3,370	**0.785 (0.771, 0.798)**
Low-evidence alert triggered	327	3,370	0.097 (0.087, 0.107)
Draft-validation failure	209	3,370	0.062 (0.054, 0.071)
Fallback after generation or formatting failure	13	3,370	0.004 (0.002, 0.007)
Physician review or escalation	807	3,370	0.239 (0.225, 0.254)
False negatives without escalation flags	23	3,370	0.007 (0.005, 0.010)

Supplementary Table S10 | Attribution of false-negative and false-positive errors in the full Qwen3-Max system

Each error category uses the total number of the corresponding error type in that cohort as the denominator. Attribution categories are other or unlocalized, venous-family-related, multiple-family combinations and substantial field missingness. Results refer to the full ExpEvaFCHD system using Qwen3-Max as the generation backend.

Prospective cohort

Error type	Attribution	Error count	Denominator	Proportion (95% CI)
FN	Other or unlocalized	11	26	0.423 (0.255, 0.611)
FN	Venous-family-related	0	26	0.000 (0.000, 0.129)
FN	Multiple-family combinations	6	26	0.231 (0.110, 0.421)
FN	Substantial field missingness	9	26	0.346 (0.194, 0.538)
FP	Other or unlocalized	8	12	0.667 (0.391, 0.862)
FP	Venous-family-related	0	12	0.000 (0.000, 0.242)
FP	Multiple-family combinations	0	12	0.000 (0.000, 0.242)
FP	Substantial field missingness	4	12	0.333 (0.138, 0.609)

Multicenter cohort

Error type	Attribution	Error count	Denominator	Proportion (95% CI)
FN	Other or unlocalized	13	33	0.394 (0.247, 0.563)
FN	Venous-family-related	0	33	0.000 (0.000, 0.104)
FN	Multiple-family combinations	5	33	0.152 (0.067, 0.309)
FN	Substantial field missingness	15	33	0.455 (0.298, 0.620)
FP	Other or unlocalized	30	58	0.517 (0.392, 0.641)
FP	Venous-family-related	0	58	0.000 (0.000, 0.062)
FP	Multiple-family combinations	0	58	0.000 (0.000, 0.062)
FP	Substantial field missingness	28	58	0.483 (0.359, 0.608)

Supplementary Table S11 | Evidence support for abnormality statements and report coverage

Physicians audited abnormality statements in model-generated reports during daily final review. SFCR is the proportion of all abnormality statements supported by evidence; UASR is the proportion lacking evidence support. Report coverage is the proportion of all reports containing abnormality statements. The first two measures use the number of abnormality statements as the denominator; report coverage uses the total number of reports.

A. Prospective cohort: abnormality-statement support and report coverage counts

Model	Condition	Reports, n	Report level	Abnormality-statement level		
			Reports with abnormality statements	Abnormality statements	Supported statements	Unsupported statements

Model	Condition	Reports, n	Report level	Abnormality-statement level		
			Reports with abnormality statements	Abnormality statements	Supported statements	Unsupported statements
Kimi-K2	Base	1,458	1,246	3,580	1,245	2,335
Kimi-K2	Full	**1,458**	**494**	**1,877**	**1,800**	**77**
DeepSeek-V3.2	Base	1,458	1,210	3,461	1,190	2,271
DeepSeek-V3.2	Full	**1,458**	**502**	**2,034**	**1,958**	**76**
GLM-4.5	Base	1,458	1,225	3,412	1,139	2,273
GLM-4.5	Full	**1,458**	**492**	**1,901**	**1,814**	**87**
Llama-3.1-70B	Base	1,458	1,217	3,522	1,220	2,302
Llama-3.1-70B	Full	**1,458**	**503**	**1,953**	**1,864**	**89**
Qwen3-Max	Base	1,458	1,200	3,515	1,177	2,338
Qwen3-Max	Full	**1,458**	**468**	**1,873**	**1,807**	**66**
GPT-5	Base	1,458	1,235	3,545	1,197	2,348
GPT-5	Full	**1,458**	**503**	**1,911**	**1,829**	**82**

B. Prospective cohort: supported and unsupported abnormality-statement rates and report coverage (95% CI)

Model	Condition	Abnormality-statement level		Report level
		SFCR (95% CI)	UASR (95% CI)	Report coverage (95% CI)
Kimi-K2	Base	34.78% (33.19, 36.28)	65.22% (63.72, 66.81)	85.46% (83.56, 87.18)
Kimi-K2	Full	**95.90% (95.02, 96.71)**	**4.10% (3.29, 4.98)**	**33.88% (31.50, 36.35)**
DeepSeek-V3.2	Base	34.38% (32.77, 36.02)	65.62% (63.98, 67.23)	82.99% (80.98, 84.83)
DeepSeek-V3.2	Full	**96.26% (95.37, 97.10)**	**3.74% (2.90, 4.63)**	**34.43% (32.04, 36.91)**
GLM-4.5	Base	33.38% (31.82, 34.94)	66.62% (65.06, 68.18)	84.02% (82.05, 85.81)
GLM-4.5	Full	**95.42% (94.51, 96.34)**	**4.58% (3.66, 5.49)**	**33.74% (31.36, 36.21)**
Llama-3.1-70B	Base	34.64% (33.04, 36.26)	65.36% (63.74, 66.96)	83.47% (81.48, 85.29)
Llama-3.1-70B	Full	**95.44% (94.50, 96.36)**	**4.56% (3.64, 5.50)**	**34.50% (32.10, 36.98)**
Qwen3-Max	Base	33.49% (31.99, 35.12)	66.51% (64.88, 68.01)	82.30% (80.26, 84.18)
Qwen3-Max	Full	**96.48% (95.64, 97.31)**	**3.52% (2.69, 4.36)**	**32.10% (29.75, 34.54)**
GPT-5	Base	33.77% (32.36, 35.27)	66.23% (64.73, 67.64)	84.71% (82.77, 86.46)
GPT-5	Full	**95.71% (94.81, 96.50)**	**4.29% (3.50, 5.19)**	**34.50% (32.10, 36.98)**

C. Multicenter cohort: abnormality-statement support and report coverage counts

Model	Condition	Reports, n	Report level	Abnormality-statement level		
			Reports with abnormality statements	Abnormality statements	Supported statements	Unsupported statements
Kimi-K2	Base	3,370	2,690	6,206	2,056	4,150
Kimi-K2	Full	**3,370**	**720**	**2,138**	**2,058**	**80**
DeepSeek-V3.2	Base	3,370	2,686	6,326	2,144	4,182
DeepSeek-V3.2	Full	**3,370**	**706**	**2,070**	**1,989**	**81**
GLM-4.5	Base	3,370	2,695	6,225	2,052	4,173
GLM-4.5	Full	**3,370**	**776**	**2,071**	**1,975**	**96**
Llama-3.1-70B	Base	3,370	2,684	6,168	2,108	4,060
Llama-3.1-70B	Full	**3,370**	**730**	**2,154**	**2,065**	**89**
Qwen3-Max	Base	3,370	2,685	6,107	2,126	3,981
Qwen3-Max	Full	**3,370**	**754**	**2,162**	**2,085**	**77**
GPT-5	Base	3,370	2,713	6,340	2,157	4,183
GPT-5	Full	**3,370**	**744**	**2,137**	**2,046**	**91**

D. Multicenter cohort: supported and unsupported abnormality-statement rates and report coverage (95% CI)

Model	Condition	Abnormality-statement level		Report level
		SFCR (95% CI)	UASR (95% CI)	Report coverage (95% CI)
Kimi-K2	Base	33.13% (31.96, 34.32)	66.87% (65.68, 68.04)	79.82% (78.43, 81.14)
Kimi-K2	Full	**96.26% (95.48, 97.01)**	**3.74% (2.99, 4.52)**	**21.36% (20.01, 22.78)**
DeepSeek-V3.2	Base	33.89% (32.66, 35.09)	66.11% (64.91, 67.34)	79.70% (78.31, 81.03)
DeepSeek-V3.2	Full	**96.09% (95.26, 96.88)**	**3.91% (3.12, 4.74)**	**20.95% (19.61, 22.36)**
GLM-4.5	Base	32.96% (31.79, 34.06)	67.04% (65.94, 68.21)	79.97% (78.59, 81.29)
GLM-4.5	Full	**95.36% (94.44, 96.18)**	**4.64% (3.82, 5.56)**	**23.03% (21.64, 24.48)**
Llama-3.1-70B	Base	34.18% (32.99, 35.32)	65.82% (64.68, 67.01)	79.64% (78.25, 80.97)
Llama-3.1-70B	Full	**95.87% (95.03, 96.64)**	**4.13% (3.36, 4.97)**	**21.66% (20.30, 23.08)**
Qwen3-Max	Base	34.81% (33.63, 36.01)	65.19% (63.99, 66.37)	79.67% (78.28, 81.00)
Qwen3-Max	Full	**96.44% (95.63, 97.18)**	**3.56% (2.82, 4.37)**	**22.37% (21.00, 23.81)**
GPT-5	Base	34.02% (32.82, 35.13)	65.98% (64.87, 67.18)	80.50% (79.13, 81.81)
GPT-5	Full	**95.74% (94.88, 96.55)**	**4.26% (3.45, 5.12)**	**22.08% (20.71, 23.51)**

Supplementary Table S12 | Binary screening results for six base models and the full system

A and C show confusion-matrix counts for the prospective and multicenter cohorts, respectively; B and D show the seven screening metrics with 95% confidence intervals for the respective cohorts.

A. Prospective cohort: confusion-matrix counts for base models and the full system

Model	Condition	Binary confusion matrix · Case counts			
		TP	FP	TN	FN
Kimi-K2	Base	204	237	881	136
Kimi-K2	Full	**312**	**17**	**1,101**	**28**
DeepSeek-V3.2	Base	249	842	276	91
DeepSeek-V3.2	Full	**311**	**15**	**1,103**	**29**
GLM-4.5	Base	221	313	805	119
GLM-4.5	Full	**308**	**20**	**1,098**	**32**
Llama-3.1-70B	Base	204	315	803	136
Llama-3.1-70B	Full	**302**	**22**	**1,096**	**38**
Qwen3-Max	Base	181	276	842	159
Qwen3-Max	Full	**314**	**12**	**1,106**	**26**
GPT-5	Base	238	224	894	102
GPT-5	Full	**316**	**13**	**1,105**	**24**

B. Prospective cohort: screening performance of base models and the full system with 95% confidence intervals

Model	Metric	Base · Direct generation		Full · ExpEvaFCHD	
		Base (95% CI)	Base n / N	Full (95% CI)	Full n / N
Kimi-K2	Sensitivity	60.00% (54.71, 65.07)	204 / 340	91.76% (88.36, 94.24)	312 / 340
Kimi-K2	Specificity	78.80% (76.31, 81.10)	881 / 1,118	98.48% (97.58, 99.05)	1,101 / 1,118
Kimi-K2	PPV	46.26% (41.66, 50.92)	204 / 441	94.83% (91.88, 96.75)	312 / 329
Kimi-K2	NPV	86.63% (84.40, 88.58)	881 / 1,017	97.52% (96.44, 98.28)	1,101 / 1,129
Kimi-K2	Accuracy	74.42% (72.12, 76.59)	1,085 / 1,458	96.91% (95.90, 97.69)	1,413 / 1,458
Kimi-K2	LR+	2.8304 (2.4545, 3.2638)	204 / 340	60.3488 (37.6124, 96.8291)	312 / 340
Kimi-K2	LR−	0.5076 (0.4441, 0.5802)	136 / 340	0.0836 (0.0586, 0.1193)	28 / 340
DeepSeek-V3.2	Sensitivity	73.24% (68.29, 77.66)	249 / 340	91.47% (88.02, 94.00)	311 / 340
DeepSeek-V3.2	Specificity	24.69% (22.25, 27.30)	276 / 1,118	98.66% (97.80, 99.19)	1,103 / 1,118
DeepSeek-V3.2	PPV	22.82% (20.43, 25.41)	249 / 1,091	95.40% (92.55, 97.19)	311 / 326
DeepSeek-V3.2	NPV	75.20% (70.54, 79.35)	276 / 367	97.44% (96.35, 98.21)	1,103 / 1,132
DeepSeek-V3.2	Accuracy	36.01% (33.58, 38.51)	525 / 1,458	96.98% (95.97, 97.74)	1,414 / 1,458

Model	Metric	Base · Direct generation		Full · ExpEvaFCHD	
		Base (95% CI)	Base n / N	Full (95% CI)	Full n / N
DeepSeek-V3.2	LR+	0.9724 (0.9044, 1.0455)	249 / 340	68.1761 (41.1981, 112.8201)	311 / 340
DeepSeek-V3.2	LR−	1.0842 (0.8846, 1.3288)	91 / 340	0.0865 (0.0610, 0.1225)	29 / 340
GLM-4.5	Sensitivity	65.00% (59.79, 69.88)	221 / 340	90.59% (87.02, 93.25)	308 / 340
GLM-4.5	Specificity	72.00% (69.30, 74.56)	805 / 1,118	98.21% (97.25, 98.84)	1,098 / 1,118
GLM-4.5	PPV	41.39% (37.28, 45.61)	221 / 534	93.90% (90.77, 96.02)	308 / 328
GLM-4.5	NPV	87.12% (84.81, 89.13)	805 / 924	97.17% (96.03, 97.99)	1,098 / 1,130
GLM-4.5	Accuracy	70.37% (67.98, 72.66)	1,026 / 1,458	96.43% (95.35, 97.27)	1,406 / 1,458
GLM-4.5	LR+	2.3217 (2.0548, 2.6234)	221 / 340	50.6388 (32.7547, 78.2878)	308 / 340
GLM-4.5	LR−	0.4861 (0.4186, 0.5644)	119 / 340	0.0958 (0.0689, 0.1333)	32 / 340
Llama-3.1-70B	Sensitivity	60.00% (54.71, 65.07)	204 / 340	88.82% (85.03, 91.75)	302 / 340
Llama-3.1-70B	Specificity	71.82% (69.12, 74.38)	803 / 1,118	98.03% (97.04, 98.70)	1,096 / 1,118
Llama-3.1-70B	PPV	39.31% (35.20, 43.57)	204 / 519	93.21% (89.93, 95.47)	302 / 324
Llama-3.1-70B	NPV	85.52% (83.12, 87.62)	803 / 939	96.65% (95.43, 97.55)	1,096 / 1,134
Llama-3.1-70B	Accuracy	69.07% (66.65, 71.39)	1,007 / 1,458	95.88% (94.74, 96.79)	1,398 / 1,458
Llama-3.1-70B	LR+	2.1295 (1.8743, 2.4194)	204 / 340	45.1385 (29.7934, 68.3871)	302 / 340
Llama-3.1-70B	LR−	0.5569 (0.4865, 0.6376)	136 / 340	0.1140 (0.0845, 0.1539)	38 / 340
Qwen3-Max	Sensitivity	53.24% (47.93, 58.47)	181 / 340	92.35% (89.03, 94.73)	314 / 340
Qwen3-Max	Specificity	75.31% (72.70, 77.75)	842 / 1,118	98.93% (98.13, 99.38)	1,106 / 1,118
Qwen3-Max	PPV	39.61% (35.23, 44.16)	181 / 457	96.32% (93.68, 97.88)	314 / 326
Qwen3-Max	NPV	84.12% (81.72, 86.25)	842 / 1,001	97.70% (96.66, 98.43)	1,106 / 1,132
Qwen3-Max	Accuracy	70.16% (67.77, 72.46)	1,023 / 1,458	97.39% (96.44, 98.10)	1,420 / 1,458
Qwen3-Max	LR+	2.1564 (1.8694, 2.4876)	181 / 340	86.0422 (48.9725, 151.1717)	314 / 340
Qwen3-Max	LR−	0.6209 (0.5517, 0.6989)	159 / 340	0.0773 (0.0534, 0.1118)	26 / 340
GPT-5	Sensitivity	70.00% (64.93, 74.63)	238 / 340	92.94% (89.71, 95.21)	316 / 340
GPT-5	Specificity	79.96% (77.52, 82.21)	894 / 1,118	98.84% (98.02, 99.32)	1,105 / 1,118
GPT-5	PPV	51.52% (46.96, 56.04)	238 / 462	96.05% (93.36, 97.68)	316 / 329
GPT-5	NPV	89.76% (87.72, 91.49)	894 / 996	97.87% (96.86, 98.57)	1,105 / 1,129
GPT-5	Accuracy	77.64% (75.43, 79.71)	1,132 / 1,458	97.46% (96.52, 98.15)	1,421 / 1,458
GPT-5	LR+	3.4938 (3.0488, 4.0036)	238 / 340	79.9294 (46.5219, 137.3268)	316 / 340
GPT-5	LR−	0.3752 (0.3181, 0.4425)	102 / 340	0.0714 (0.0486, 0.1050)	24 / 340

C. Multicenter cohort: confusion-matrix counts for base models and the full system

Model	Condition	Binary confusion matrix · Case counts			
		TP	FP	TN	FN
Kimi-K2	Base	180	672	2,382	136
Kimi-K2	Full	**281**	**70**	**2,984**	**35**
DeepSeek-V3.2	Base	222	2,324	730	94
DeepSeek-V3.2	Full	**280**	**64**	**2,990**	**36**
GLM-4.5	Base	196	880	2,174	120
GLM-4.5	Full	**277**	**79**	**2,975**	**39**
Llama-3.1-70B	Base	180	886	2,168	136
Llama-3.1-70B	Full	**271**	**86**	**2,968**	**45**
Qwen3-Max	Base	159	779	2,275	157
Qwen3-Max	Full	**283**	**58**	**2,996**	**33**
GPT-5	Base	212	635	2,419	104
GPT-5	Full	**284**	**61**	**2,993**	**32**

D. Multicenter cohort: screening performance of base models and the full system with 95% confidence intervals

Model	Metric	Base · Direct generation		Full · ExpEvaFCHD	
		Base (95% CI)	Base n / N	Full (95% CI)	Full n / N
Kimi-K2	Sensitivity	56.96% (51.45, 62.31)	180 / 316	88.92% (84.99, 91.93)	281 / 316
Kimi-K2	Specificity	78.00% (76.49, 79.43)	2,382 / 3,054	97.71% (97.11, 98.18)	2,984 / 3,054
Kimi-K2	PPV	21.13% (18.52, 23.99)	180 / 852	80.06% (75.56, 83.90)	281 / 351
Kimi-K2	NPV	94.60% (93.65, 95.42)	2,382 / 2,518	98.84% (98.39, 99.17)	2,984 / 3,019
Kimi-K2	Accuracy	76.02% (74.55, 77.44)	2,562 / 3,370	96.88% (96.24, 97.42)	3,265 / 3,370
Kimi-K2	LR+	2.5887 (2.3033, 2.9095)	180 / 316	38.7963 (30.6771, 49.0643)	281 / 316
Kimi-K2	LR−	0.5518 (0.4854, 0.6273)	136 / 316	0.1134 (0.0829, 0.1549)	35 / 316
DeepSeek-V3.2	Sensitivity	70.25% (64.99, 75.03)	222 / 316	88.61% (84.63, 91.66)	280 / 316
DeepSeek-V3.2	Specificity	23.90% (22.42, 25.45)	730 / 3,054	97.90% (97.33, 98.36)	2,990 / 3,054
DeepSeek-V3.2	PPV	8.72% (7.68, 9.88)	222 / 2,546	81.40% (76.94, 85.15)	280 / 344
DeepSeek-V3.2	NPV	88.59% (86.24, 90.59)	730 / 824	98.81% (98.36, 99.14)	2,990 / 3,026
DeepSeek-V3.2	Accuracy	28.25% (26.75, 29.79)	952 / 3,370	97.03% (96.40, 97.55)	3,270 / 3,370
DeepSeek-V3.2	LR+	0.9232 (0.8570, 0.9946)	222 / 316	42.2824 (33.0742, 54.0543)	280 / 316
DeepSeek-V3.2	LR−	1.2445 (1.0386, 1.4912)	94 / 316	0.1164 (0.0856, 0.1583)	36 / 316
GLM-4.5	Sensitivity	62.03% (56.56, 67.20)	196 / 316	87.66% (83.57, 90.84)	277 / 316

Model	Metric	Base · Direct generation		Full · ExpEvaFCHD	
		Base (95% CI)	Base n / N	Full (95% CI)	Full n / N
GLM-4.5	Specificity	71.19% (69.55, 72.76)	2,174 / 3,054	97.41% (96.79, 97.92)	2,975 / 3,054
GLM-4.5	PPV	18.22% (16.02, 20.63)	196 / 1,076	77.81% (73.21, 81.82)	277 / 356
GLM-4.5	NPV	94.77% (93.78, 95.61)	2,174 / 2,294	98.71% (98.24, 99.05)	2,975 / 3,014
GLM-4.5	Accuracy	70.33% (68.76, 71.84)	2,370 / 3,370	96.50% (95.82, 97.07)	3,252 / 3,370
GLM-4.5	LR+	2.1526 (1.9424, 2.3854)	196 / 316	33.8871 (27.1532, 42.2910)	277 / 316
GLM-4.5	LR−	0.5335 (0.4625, 0.6153)	120 / 316	0.1267 (0.0944, 0.1700)	39 / 316
Llama-3.1-70B	Sensitivity	56.96% (51.45, 62.31)	180 / 316	85.76% (81.48, 89.18)	271 / 316
Llama-3.1-70B	Specificity	70.99% (69.35, 72.57)	2,168 / 3,054	97.18% (96.54, 97.71)	2,968 / 3,054
Llama-3.1-70B	PPV	16.89% (14.76, 19.25)	180 / 1,066	75.91% (71.21, 80.06)	271 / 357
Llama-3.1-70B	NPV	94.10% (93.06, 94.99)	2,168 / 2,304	98.51% (98.01, 98.88)	2,968 / 3,013
Llama-3.1-70B	Accuracy	69.67% (68.10, 71.20)	2,348 / 3,370	96.11% (95.41, 96.71)	3,239 / 3,370
Llama-3.1-70B	LR+	1.9635 (1.7576, 2.1934)	180 / 316	30.4546 (24.6086, 37.6893)	271 / 316
Llama-3.1-70B	LR−	0.6063 (0.5330, 0.6896)	136 / 316	0.1465 (0.1118, 0.1921)	45 / 316
Qwen3-Max	Sensitivity	50.32% (44.83, 55.79)	159 / 316	89.56% (85.70, 92.47)	283 / 316
Qwen3-Max	Specificity	74.49% (72.92, 76.01)	2,275 / 3,054	98.10% (97.55, 98.53)	2,996 / 3,054
Qwen3-Max	PPV	16.95% (14.69, 19.49)	159 / 938	82.99% (78.64, 86.61)	283 / 341
Qwen3-Max	NPV	93.54% (92.50, 94.45)	2,275 / 2,432	98.91% (98.47, 99.22)	2,996 / 3,029
Qwen3-Max	Accuracy	72.23% (70.69, 73.71)	2,434 / 3,370	97.30% (96.70, 97.80)	3,279 / 3,370
Qwen3-Max	LR+	1.9726 (1.7405, 2.2357)	159 / 316	47.1564 (36.4450, 61.0159)	283 / 316
Qwen3-Max	LR−	0.6670 (0.5958, 0.7467)	157 / 316	0.1065 (0.0771, 0.1470)	33 / 316
GPT-5	Sensitivity	67.09% (61.73, 72.04)	212 / 316	89.87% (86.05, 92.74)	284 / 316
GPT-5	Specificity	79.21% (77.73, 80.61)	2,419 / 3,054	98.00% (97.44, 98.44)	2,993 / 3,054
GPT-5	PPV	25.03% (22.23, 28.06)	212 / 847	82.32% (77.94, 85.98)	284 / 345
GPT-5	NPV	95.88% (95.03, 96.59)	2,419 / 2,523	98.94% (98.51, 99.25)	2,993 / 3,025
GPT-5	Accuracy	78.07% (76.64, 79.44)	2,631 / 3,370	97.24% (96.63, 97.74)	3,277 / 3,370
GPT-5	LR+	3.2266 (2.9087, 3.5792)	212 / 316	44.9956 (35.0016, 57.8432)	284 / 316
GPT-5	LR−	0.4155 (0.3546, 0.4869)	104 / 316	0.1033 (0.0744, 0.1435)	32 / 316

Supplementary Table S13 | Binary screening results for comparators and component-ablation conditions

A shows confusion-matrix counts for each comparator and ablation condition in both validation cohorts; B and C show screening metrics with 95% confidence intervals for the prospective and multicenter cohorts, respectively.

A. Prospective and multicenter cohorts: confusion-matrix counts for comparators and ablation conditions

Condition	Prospective cohort · n = 1,458				Multicenter cohort · n = 3,370			
	Prospective TP	Prospective FP	Prospective TN	Prospective FN	Multicenter TP	Multicenter FP	Multicenter TN	Multicenter FN
GA rules	272	67	1,051	68	243	208	2,846	73
kNN	286	50	1,068	54	256	162	2,892	60
Logistic regression	269	61	1,057	71	240	192	2,862	76
Text RAG	265	157	961	75	237	452	2,602	79
Structured RAG no validation	316	61	1,057	24	284	192	2,862	32
No GA normalization	294	32	1,086	46	264	113	2,941	52
No polarity control	310	58	1,060	30	279	183	2,871	37
No SAWS threshold	323	48	1,070	17	291	156	2,898	25
No closure	321	65	1,053	19	289	202	2,852	27
kNN + template	286	50	1,068	54	256	162	2,892	60
Matched evidence + template	314	12	1,106	26	283	58	2,996	33
Full Qwen3-Max	314	12	1,106	26	283	58	2,996	33
Direct Qwen3-Max	181	276	842	159	159	779	2,275	157

B. Prospective cohort: screening performance of comparators and ablation conditions with 95% confidence intervals

Method / ablation condition	Metric group I				Metric group II			
	Metric	Estimate (95% CI)	Numerator n	Reference N	Metric	Estimate (95% CI)	Numerator n	Reference N
GA rules	Sensitivity	80.00% (75.42, 83.91)	272	340	Accuracy	90.74% (89.14, 92.12)	1,323	1,458
GA rules	Specificity	94.01% (92.46, 95.25)	1,051	1,118	LR+	13.3493 (10.5202, 16.9392)	272	340
GA rules	PPV	80.24% (75.67, 84.13)	272	339	LR−	0.2127 (0.1719, 0.2633)	68	340
GA rules	NPV	93.92% (92.37, 95.18)	1,051	1,119				
kNN	Sensitivity	84.12% (79.85, 87.62)	286	340	Accuracy	92.87% (91.43, 94.08)	1,354	1,458
kNN	Specificity	95.53% (94.15, 96.59)	1,068	1,118	LR+	18.8087 (14.2891, 24.7578)	286	340
kNN	PPV	85.12% (80.92, 88.53)	286	336	LR−	0.1663 (0.1301, 0.2124)	54	340
kNN	NPV	95.19% (93.77, 96.29)	1,068	1,122				
Logistic regression	Sensitivity	79.12% (74.48, 83.10)	269	340	Accuracy	90.95% (89.36, 92.31)	1,326	1,458
Logistic regression	Specificity	94.54% (93.05, 95.73)	1,057	1,118	LR+	14.5006 (11.2926, 18.6199)	269	340

Method / ablation condition	Metric group I				Metric group II			
	Metric	Estimate (95% CI)	Numerator n	Reference N	Metric	Estimate (95% CI)	Numerator n	Reference N
Logistic regression	PPV	81.52% (76.97, 85.33)	269	330	LR−	0.2209 (0.1795, 0.2718)	71	340
Logistic regression	NPV	93.71% (92.13, 94.98)	1,057	1,128				
Text RAG	Sensitivity	77.94% (73.24, 82.02)	265	340	Accuracy	84.09% (82.12, 85.88)	1,226	1,458
Text RAG	Specificity	85.96% (83.80, 87.87)	961	1,118	LR+	5.5502 (4.7502, 6.4850)	265	340
Text RAG	PPV	62.80% (58.09, 67.27)	265	422	LR−	0.2566 (0.2099, 0.3138)	75	340
Text RAG	NPV	92.76% (91.02, 94.19)	961	1,036				
Structured RAG no validation	Sensitivity	92.94% (89.71, 95.21)	316	340	Accuracy	94.17% (92.85, 95.26)	1,373	1,458
Structured RAG no validation	Specificity	94.54% (93.05, 95.73)	1,057	1,118	LR+	17.0341 (13.3226, 21.7797)	316	340
Structured RAG no validation	PPV	83.82% (79.76, 87.19)	316	377	LR−	0.0747 (0.0508, 0.1098)	24	340
Structured RAG no validation	NPV	97.78% (96.72, 98.50)	1,057	1,081				
No GA normalization	Sensitivity	86.47% (82.42, 89.70)	294	340	Accuracy	94.65% (93.37, 95.69)	1,380	1,458
No GA normalization	Specificity	97.14% (95.99, 97.97)	1,086	1,118	LR+	30.2107 (21.4159, 42.6171)	294	340
No GA normalization	PPV	90.18% (86.47, 92.96)	294	326	LR−	0.1393 (0.1064, 0.1823)	46	340
No GA normalization	NPV	95.94% (94.62, 96.94)	1,086	1,132				
No polarity control	Sensitivity	91.18% (87.68, 93.75)	310	340	Accuracy	93.96% (92.62, 95.08)	1,370	1,458
No polarity control	Specificity	94.81% (93.35, 95.97)	1,060	1,118	LR+	17.5751 (13.6497, 22.6293)	310	340
No polarity control	PPV	84.24% (80.17, 87.61)	310	368	LR−	0.0931 (0.0661, 0.1310)	30	340
No polarity control	NPV	97.25% (96.10, 98.07)	1,060	1,090				
No SAWS threshold	Sensitivity	95.00% (92.14, 96.86)	323	340	Accuracy	95.54% (94.36, 96.49)	1,393	1,458
No SAWS threshold	Specificity	95.71% (94.35, 96.75)	1,070	1,118	LR+	22.1271 (16.7596, 29.2135)	323	340
No SAWS threshold	PPV	87.06% (83.26, 90.10)	323	371	LR−	0.0522 (0.0329, 0.0830)	17	340
No SAWS threshold	NPV	98.44% (97.51, 99.02)	1,070	1,087				
No closure	Sensitivity	94.41% (91.44, 96.39)	321	340	Accuracy	94.24% (92.92, 95.32)	1,374	1,458
No closure	Specificity	94.19% (92.66, 95.41)	1,053	1,118	LR+	16.2388 (12.8079, 20.5888)	321	340
No closure	PPV	83.16% (79.10, 86.56)	321	386	LR−	0.0593 (0.0383, 0.0919)	19	340
No closure	NPV	98.23% (97.25, 98.86)	1,053	1,072				
kNN + template	Sensitivity	84.12% (79.85, 87.62)	286	340	Accuracy	92.87% (91.43, 94.08)	1,354	1,458
kNN + template	Specificity	95.53% (94.15, 96.59)	1,068	1,118	LR+	18.8087 (14.2891, 24.7578)	286	340
kNN + template	PPV	85.12% (80.92, 88.53)	286	336	LR−	0.1663 (0.1301, 0.2124)	54	340
kNN + template	NPV	95.19% (93.77, 96.29)	1,068	1,122				
Matched evidence + template	Sensitivity	92.35% (89.03, 94.73)	314	340	Accuracy	97.39% (96.44, 98.10)	1,420	1,458
Matched evidence + template	Specificity	98.93% (98.13, 99.38)	1,106	1,118	LR+	86.0422 (48.9725, 151.1717)	314	340

Method / ablation condition	Metric group I				Metric group II			
	Metric	Estimate (95% CI)	Numerator n	Reference N	Metric	Estimate (95% CI)	Numerator n	Reference N
Matched evidence + template	PPV	96.32% (93.68, 97.88)	314	326	LR−	0.0773 (0.0534, 0.1118)	26	340
Matched evidence + template	NPV	97.70% (96.66, 98.43)	1,106	1,132				
Full Qwen3-Max	Sensitivity	**92.35% (89.03, 94.73)**	314	340	Accuracy	**97.39% (96.44, 98.10)**	1,420	1,458
Full Qwen3-Max	Specificity	**98.93% (98.13, 99.38)**	1,106	1,118	LR+	**86.0422 (48.9725, 151.1717)**	314	340
Full Qwen3-Max	PPV	**96.32% (93.68, 97.88)**	314	326	LR−	**0.0773 (0.0534, 0.1118)**	26	340
Full Qwen3-Max	NPV	**97.70% (96.66, 98.43)**	1,106	1,132				
Direct Qwen3-Max	Sensitivity	53.24% (47.93, 58.47)	181	340	Accuracy	70.16% (67.77, 72.46)	1,023	1,458
Direct Qwen3-Max	Specificity	75.31% (72.70, 77.75)	842	1,118	LR+	2.1564 (1.8694, 2.4876)	181	340
Direct Qwen3-Max	PPV	39.61% (35.23, 44.16)	181	457	LR−	0.6209 (0.5517, 0.6989)	159	340
Direct Qwen3-Max	NPV	84.12% (81.72, 86.25)	842	1,001				

C. Multicenter cohort: screening performance of comparators and ablation conditions with 95% confidence intervals

Method / ablation condition	Metric group I				Metric group II			
	Metric	Estimate (95% CI)	Numerator n	Reference N	Metric	Estimate (95% CI)	Numerator n	Reference N
GA rules	Sensitivity	76.90% (71.95, 81.21)	243	316	Accuracy	91.66% (90.68, 92.55)	3,089	3,370
GA rules	Specificity	93.19% (92.24, 94.03)	2,846	3,054	LR+	11.2908 (9.7723, 13.0453)	243	316
GA rules	PPV	53.88% (49.27, 58.43)	243	451	LR−	0.2479 (0.2027, 0.3032)	73	316
GA rules	NPV	97.50% (96.87, 98.01)	2,846	2,919				
kNN	Sensitivity	81.01% (76.33, 84.95)	256	316	Accuracy	93.41% (92.52, 94.20)	3,148	3,370
kNN	Specificity	94.70% (93.84, 95.44)	2,892	3,054	LR+	15.2724 (13.0263, 17.9057)	256	316
kNN	PPV	61.24% (56.49, 65.79)	256	418	LR−	0.2005 (0.1596, 0.2518)	60	316
kNN	NPV	97.97% (97.39, 98.42)	2,892	2,952				
Logistic regression	Sensitivity	75.95% (70.94, 80.33)	240	316	Accuracy	92.05% (91.09, 92.91)	3,102	3,370
Logistic regression	Specificity	93.71% (92.80, 94.52)	2,862	3,054	LR+	12.0807 (10.3945, 14.0404)	240	316
Logistic regression	PPV	55.56% (50.84, 60.17)	240	432	LR−	0.2566 (0.2109, 0.3123)	76	316
Logistic regression	NPV	97.41% (96.77, 97.93)	2,862	2,938				
Text RAG	Sensitivity	75.00% (69.94, 79.45)	237	316	Accuracy	84.24% (82.97, 85.43)	2,839	3,370
Text RAG	Specificity	85.20% (83.90, 86.41)	2,602	3,054	LR+	5.0675 (4.5566, 5.6356)	237	316
Text RAG	PPV	34.40% (30.95, 38.02)	237	689	LR−	0.2934 (0.2423, 0.3554)	79	316
Text RAG	NPV	97.05% (96.34, 97.63)	2,602	2,681				
Structured RAG no validation	Sensitivity	89.87% (86.05, 92.74)	284	316	Accuracy	93.35% (92.46, 94.15)	3,146	3,370

Method / ablation condition	Metric group I				Metric group II			
	Metric	Estimate (95% CI)	Numerator n	Reference N	Metric	Estimate (95% CI)	Numerator n	Reference N
Structured RAG no validation	Specificity	93.71% (92.80, 94.52)	2,862	3,054	LR+	14.2955 (12.4050, 16.4741)	284	316
Structured RAG no validation	PPV	59.66% (55.20, 63.98)	284	476	LR−	0.1081 (0.0778, 0.1501)	32	316
Structured RAG no validation	NPV	98.89% (98.44, 99.22)	2,862	2,894				
No GA normalization	Sensitivity	83.54% (79.06, 87.22)	264	316	Accuracy	95.10% (94.32, 95.78)	3,205	3,370
No GA normalization	Specificity	96.30% (95.57, 96.91)	2,941	3,054	LR+	22.5791 (18.7200, 27.2339)	264	316
No GA normalization	PPV	70.03% (65.22, 74.43)	264	377	LR−	0.1709 (0.1333, 0.2191)	52	316
No GA normalization	NPV	98.26% (97.73, 98.67)	2,941	2,993				
No polarity control	Sensitivity	88.29% (84.28, 91.38)	279	316	Accuracy	93.47% (92.59, 94.26)	3,150	3,370
No polarity control	Specificity	94.01% (93.11, 94.80)	2,871	3,054	LR+	14.7345 (12.7316, 17.0524)	279	316
No polarity control	PPV	60.39% (55.86, 64.75)	279	462	LR−	0.1246 (0.0920, 0.1686)	37	316
No polarity control	NPV	98.73% (98.25, 99.08)	2,871	2,908				
No SAWS threshold	Sensitivity	92.09% (88.58, 94.58)	291	316	Accuracy	94.63% (93.82, 95.34)	3,189	3,370
No SAWS threshold	Specificity	94.89% (94.05, 95.62)	2,898	3,054	LR+	18.0281 (15.4204, 21.0768)	291	316
No SAWS threshold	PPV	65.10% (60.57, 69.37)	291	447	LR−	0.0834 (0.0572, 0.1215)	25	316
No SAWS threshold	NPV	99.14% (98.74, 99.42)	2,898	2,923				
No closure	Sensitivity	91.46% (87.85, 94.06)	289	316	Accuracy	93.20% (92.30, 94.01)	3,141	3,370
No closure	Specificity	93.39% (92.45, 94.21)	2,852	3,054	LR+	13.8270 (12.0512, 15.8645)	289	316
No closure	PPV	58.86% (54.45, 63.13)	289	491	LR−	0.0915 (0.0638, 0.1313)	27	316
No closure	NPV	99.06% (98.64, 99.35)	2,852	2,879				
kNN + template	Sensitivity	81.01% (76.33, 84.95)	256	316	Accuracy	93.41% (92.52, 94.20)	3,148	3,370
kNN + template	Specificity	94.70% (93.84, 95.44)	2,892	3,054	LR+	15.2724 (13.0263, 17.9057)	256	316
kNN + template	PPV	61.24% (56.49, 65.79)	256	418	LR−	0.2005 (0.1596, 0.2518)	60	316
kNN + template	NPV	97.97% (97.39, 98.42)	2,892	2,952				
Matched evidence + template	Sensitivity	89.56% (85.70, 92.47)	283	316	Accuracy	97.30% (96.70, 97.80)	3,279	3,370
Matched evidence + template	Specificity	98.10% (97.55, 98.53)	2,996	3,054	LR+	47.1564 (36.4450, 61.0159)	283	316
Matched evidence + template	PPV	82.99% (78.64, 86.61)	283	341	LR−	0.1065 (0.0771, 0.1470)	33	316
Matched evidence + template	NPV	98.91% (98.47, 99.22)	2,996	3,029				
Full Qwen3-Max	Sensitivity	**89.56% (85.70, 92.47)**	283	316	Accuracy	**97.30% (96.70, 97.80)**	3,279	3,370
Full Qwen3-Max	Specificity	**98.10% (97.55, 98.53)**	2,996	3,054	LR+	**47.1564 (36.4450, 61.0159)**	283	316
Full Qwen3-Max	PPV	**82.99% (78.64, 86.61)**	283	341	LR−	**0.1065 (0.0771, 0.1470)**	33	316
Full Qwen3-Max	NPV	**98.91% (98.47, 99.22)**	2,996	3,029				
Direct Qwen3-Max	Sensitivity	50.32% (44.83, 55.79)	159	316	Accuracy	72.23% (70.69, 73.71)	2,434	3,370

Method / ablation condition	Metric group I				Metric group II			
	Metric	Estimate (95% CI)	Numerator n	Reference N	Metric	Estimate (95% CI)	Numerator n	Reference N
Direct Qwen3-Max	Specificity	74.49% (72.92, 76.01)	2,275	3,054	LR+	1.9726 (1.7405, 2.2357)	159	316
Direct Qwen3-Max	PPV	16.95% (14.69, 19.49)	159	938	LR−	0.6670 (0.5958, 0.7467)	157	316
Direct Qwen3-Max	NPV	93.54% (92.50, 94.45)	2,275	2,432				

Supplementary Table S14 | Screening performance of six models across five provinces in the multicenter cohort

Case counts, reference-positive counts, confusion-matrix counts and screening metrics are reported separately for each province. Provincial case and positive-case counts apply to all models and conditions within that province.

Fujian: n = 1,691; reference-positive cases = 59

Model	Condition	TP / FP / TN / FN	Screening sensitivity and specificity		Predictive values	
			Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)
Kimi-K2	Base	36 / 363 / 1,269 / 23	61.02% (48.27, 72.42)	77.76% (75.68, 79.71)	9.02% (6.59, 12.24)	98.22% (97.34, 98.81)
Kimi-K2	Full	**54 / 41 / 1,591 / 5**	**91.53% (81.65, 96.33)**	**97.49% (96.61, 98.14)**	**56.84% (46.81, 66.34)**	**99.69% (99.27, 99.87)**
DeepSeek-V3.2	Base	42 / 1,231 / 401 / 17	71.19% (58.62, 81.16)	24.57% (22.54, 26.72)	3.30% (2.45, 4.43)	95.93% (93.58, 97.45)
DeepSeek-V3.2	Full	**51 / 37 / 1,595 / 8**	**86.44% (75.46, 92.97)**	**97.73% (96.89, 98.35)**	**57.95% (47.52, 67.72)**	**99.50% (99.02, 99.75)**
GLM-4.5	Base	37 / 475 / 1,157 / 22	62.71% (49.95, 73.92)	70.89% (68.64, 73.05)	7.23% (5.29, 9.80)	98.13% (97.19, 98.76)
GLM-4.5	Full	**51 / 42 / 1,590 / 8**	**86.44% (75.46, 92.97)**	**97.43% (96.54, 98.09)**	**54.84% (44.73, 64.56)**	**99.50% (99.02, 99.75)**
Llama-3.1-70B	Base	39 / 470 / 1,162 / 20	66.10% (53.37, 76.86)	71.20% (68.96, 73.35)	7.66% (5.66, 10.30)	98.31% (97.40, 98.90)
Llama-3.1-70B	Full	**51 / 44 / 1,588 / 8**	**86.44% (75.46, 92.97)**	**97.30% (96.40, 97.99)**	**53.68% (43.71, 63.37)**	**99.50% (99.01, 99.75)**
Qwen3-Max	Base	28 / 434 / 1,198 / 31	47.46% (35.27, 59.96)	73.41% (71.21, 75.49)	6.06% (4.23, 8.62)	97.48% (96.44, 98.22)
Qwen3-Max	Full	**51 / 35 / 1,597 / 8**	**86.44% (75.46, 92.97)**	**97.86% (97.03, 98.45)**	**59.30% (48.74, 69.07)**	**99.50% (99.02, 99.75)**
GPT-5	Base	39 / 349 / 1,283 / 20	66.10% (53.37, 76.86)	78.62% (76.56, 80.54)	10.05% (7.44, 13.45)	98.47% (97.64, 99.00)
GPT-5	Full	**53 / 34 / 1,598 / 6**	**89.83% (79.54, 95.26)**	**97.92% (97.10, 98.51)**	**60.92% (50.41, 70.50)**	**99.63% (99.19, 99.83)**

Ningxia: n = 899; reference-positive cases = 64

Model	Condition	TP / FP / TN / FN	Screening sensitivity and specificity		Predictive values	
			Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)

Model	Condition	TP / FP / TN / FN	Screening sensitivity and specificity		Predictive values	
			Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)
Kimi-K2	Base	30 / 192 / 643 / 34	46.88% (35.18, 58.93)	77.01% (74.03, 79.73)	13.51% (9.63, 18.64)	94.98% (93.06, 96.38)
Kimi-K2	Full	**53 / 17 / 818 / 11**	**82.81% (71.79, 90.12)**	**97.96% (96.76, 98.73)**	**75.71% (64.50, 84.25)**	**98.67% (97.64, 99.26)**
DeepSeek-V3.2	Base	38 / 643 / 192 / 26	59.38% (47.15, 70.54)	22.99% (20.27, 25.97)	5.58% (4.09, 7.57)	88.07% (83.10, 91.73)
DeepSeek-V3.2	Full	**56 / 13 / 822 / 8**	**87.50% (77.23, 93.53)**	**98.44% (97.35, 99.09)**	**81.16% (70.39, 88.65)**	**99.04% (98.11, 99.51)**
GLM-4.5	Base	36 / 236 / 599 / 28	56.25% (44.09, 67.71)	71.74% (68.59, 74.69)	13.24% (9.72, 17.78)	95.53% (93.62, 96.89)
GLM-4.5	Full	**52 / 19 / 816 / 12**	**81.25% (70.03, 88.94)**	**97.72% (96.47, 98.54)**	**73.24% (61.95, 82.15)**	**98.55% (97.48, 99.17)**
Llama-3.1-70B	Base	31 / 251 / 584 / 33	48.44% (36.63, 60.42)	69.94% (66.74, 72.95)	10.99% (7.85, 15.18)	94.65% (92.58, 96.17)
Llama-3.1-70B	Full	**52 / 24 / 811 / 12**	**81.25% (70.03, 88.94)**	**97.13% (95.76, 98.06)**	**68.42% (57.30, 77.77)**	**98.54% (97.47, 99.16)**
Qwen3-Max	Base	27 / 207 / 628 / 37	42.19% (30.87, 54.39)	75.21% (72.17, 78.02)	11.54% (8.05, 16.27)	94.44% (92.43, 95.94)
Qwen3-Max	Full	**55 / 12 / 823 / 9**	**85.94% (75.38, 92.42)**	**98.56% (97.50, 99.18)**	**82.09% (71.25, 89.45)**	**98.92% (97.96, 99.43)**
GPT-5	Base	34 / 169 / 666 / 30	53.13% (41.07, 64.82)	79.76% (76.90, 82.35)	16.75% (12.24, 22.49)	95.69% (93.91, 96.96)
GPT-5	Full	**57 / 12 / 823 / 7**	**89.06% (79.10, 94.60)**	**98.56% (97.50, 99.18)**	**82.61% (72.02, 89.76)**	**99.16% (98.27, 99.59)**

Guangdong: n = 592; reference-positive cases = 144

Model	Condition	TP / FP / TN / FN	Screening sensitivity and specificity		Predictive values	
			Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)
Kimi-K2	Base	81 / 92 / 356 / 63	56.25% (48.09, 64.09)	79.46% (75.48, 82.95)	46.82% (39.54, 54.24)	84.96% (81.23, 88.07)
Kimi-K2	Full	**129 / 10 / 438 / 15**	**89.58% (83.52, 93.59)**	**97.77% (95.94, 98.78)**	**92.81% (87.26, 96.05)**	**96.69% (94.61, 97.98)**
DeepSeek-V3.2	Base	103 / 343 / 105 / 41	71.53% (63.67, 78.26)	23.44% (19.75, 27.58)	23.09% (19.42, 27.23)	71.92% (64.14, 78.57)
DeepSeek-V3.2	Full	**129 / 12 / 436 / 15**	**89.58% (83.52, 93.59)**	**97.32% (95.38, 98.46)**	**91.49% (85.71, 95.06)**	**96.67% (94.59, 97.97)**
GLM-4.5	Base	89 / 132 / 316 / 55	61.81% (53.66, 69.34)	70.54% (66.15, 74.57)	40.27% (34.02, 46.85)	85.18% (81.20, 88.43)
GLM-4.5	Full	**128 / 12 / 436 / 16**	**88.89% (82.71, 93.04)**	**97.32% (95.38, 98.46)**	**91.43% (85.62, 95.03)**	**96.46% (94.33, 97.81)**
Llama-3.1-70B	Base	79 / 129 / 319 / 65	54.86% (46.71, 62.76)	71.21% (66.85, 75.20)	37.98% (31.66, 44.74)	83.07% (79.00, 86.49)
Llama-3.1-70B	Full	**124 / 12 / 436 / 20**	**86.11% (79.52, 90.83)**	**97.32% (95.38, 98.46)**	**91.18% (85.21, 94.88)**	**95.61% (93.32, 97.14)**
Qwen3-Max	Base	76 / 110 / 338 / 68	52.78% (44.66, 60.75)	75.45% (71.26, 79.20)	40.86% (34.05, 48.04)	83.25% (79.31, 86.57)
Qwen3-Max	Full	**130 / 9 / 439 / 14**	**90.28% (84.34, 94.12)**	**97.99% (96.23, 98.94)**	**93.53% (88.15, 96.56)**	**96.91% (94.88, 98.15)**
GPT-5	Base	103 / 85 / 363 / 41	71.53% (63.67, 78.26)	81.03% (77.14, 84.39)	54.79% (47.65, 61.73)	89.85% (86.52, 92.43)
GPT-5	Full	**130 / 8 / 440 / 14**	**90.28% (84.34, 94.12)**	**98.21% (96.52, 99.09)**	**94.20% (88.98, 97.03)**	**96.92% (94.89, 98.15)**

Anhui: n = 96; reference-positive cases = 47

Model	Condition	TP / FP / TN / FN	Screening sensitivity and specificity		Predictive values	
			Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)
Kimi-K2	Base	31 / 8 / 41 / 16	65.96% (51.67, 77.83)	83.67% (70.96, 91.49)	79.49% (64.47, 89.22)	71.93% (59.17, 81.92)
Kimi-K2	Full	**43 / 0 / 49 / 4**	**91.49% (80.07, 96.64)**	**100.00% (92.73, 100.00)**	**100.00% (91.80, 100.00)**	**92.45% (82.14, 97.03)**
DeepSeek-V3.2	Base	37 / 40 / 9 / 10	78.72% (65.10, 88.01)	18.37% (9.98, 31.36)	48.05% (37.25, 59.04)	47.37% (27.33, 68.29)
DeepSeek-V3.2	Full	**42 / 0 / 49 / 5**	**89.36% (77.41, 95.37)**	**100.00% (92.73, 100.00)**	**100.00% (91.62, 100.00)**	**90.74% (80.09, 95.98)**
GLM-4.5	Base	32 / 10 / 39 / 15	68.09% (53.83, 79.60)	79.59% (66.36, 88.52)	76.19% (61.47, 86.52)	72.22% (59.11, 82.38)
GLM-4.5	Full	**44 / 0 / 49 / 3**	**93.62% (82.84, 97.81)**	**100.00% (92.73, 100.00)**	**100.00% (91.97, 100.00)**	**94.23% (84.36, 98.02)**
Llama-3.1-70B	Base	30 / 9 / 40 / 17	63.83% (49.54, 76.03)	81.63% (68.64, 90.02)	76.92% (61.66, 87.35)	70.18% (57.34, 80.47)
Llama-3.1-70B	Full	**42 / 1 / 48 / 5**	**89.36% (77.41, 95.37)**	**97.96% (89.31, 99.64)**	**97.67% (87.94, 99.59)**	**90.57% (79.75, 95.90)**
Qwen3-Max	Base	26 / 11 / 38 / 21	55.32% (41.25, 68.59)	77.55% (64.12, 86.98)	70.27% (54.22, 82.51)	64.41% (51.66, 75.40)
Qwen3-Max	Full	**45 / 1 / 48 / 2**	**95.74% (85.75, 98.83)**	**97.96% (89.31, 99.64)**	**97.83% (88.66, 99.62)**	**96.00% (86.54, 98.90)**
GPT-5	Base	34 / 10 / 39 / 13	72.34% (58.24, 83.06)	79.59% (66.36, 88.52)	77.27% (63.01, 87.16)	75.00% (61.79, 84.77)
GPT-5	Full	**42 / 0 / 49 / 5**	**89.36% (77.41, 95.37)**	**100.00% (92.73, 100.00)**	**100.00% (91.62, 100.00)**	**90.74% (80.09, 95.98)**

Liaoning: n = 92; reference-positive cases = 2

Model	Condition	TP / FP / TN / FN	Screening sensitivity and specificity		Predictive values	
			Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)
Kimi-K2	Base	2 / 17 / 73 / 0	100.00% (34.24, 100.00)	81.11% (71.82, 87.86)	10.53% (2.94, 31.39)	100.00% (95.00, 100.00)
Kimi-K2	Full	**2 / 2 / 88 / 0**	**100.00% (34.24, 100.00)**	**97.78% (92.26, 99.39)**	**50.00% (15.00, 85.00)**	**100.00% (95.82, 100.00)**
DeepSeek-V3.2	Base	2 / 67 / 23 / 0	100.00% (34.24, 100.00)	25.56% (17.67, 35.44)	2.90% (0.80, 9.97)	100.00% (85.69, 100.00)
DeepSeek-V3.2	Full	**2 / 2 / 88 / 0**	**100.00% (34.24, 100.00)**	**97.78% (92.26, 99.39)**	**50.00% (15.00, 85.00)**	**100.00% (95.82, 100.00)**
GLM-4.5	Base	2 / 27 / 63 / 0	100.00% (34.24, 100.00)	70.00% (59.87, 78.49)	6.90% (1.91, 21.96)	100.00% (94.25, 100.00)
GLM-4.5	Full	**2 / 6 / 84 / 0**	**100.00% (34.24, 100.00)**	**93.33% (86.21, 96.91)**	**25.00% (7.15, 59.07)**	**100.00% (95.63, 100.00)**
Llama-3.1-70B	Base	1 / 27 / 63 / 1	50.00% (9.45, 90.55)	70.00% (59.87, 78.49)	3.57% (0.63, 17.71)	98.44% (91.67, 99.72)
Llama-3.1-70B	Full	**2 / 5 / 85 / 0**	**100.00% (34.24, 100.00)**	**94.44% (87.65, 97.60)**	**28.57% (8.22, 64.11)**	**100.00% (95.68, 100.00)**
Qwen3-Max	Base	2 / 17 / 73 / 0	100.00% (34.24, 100.00)	81.11% (71.82, 87.86)	10.53% (2.94, 31.39)	100.00% (95.00, 100.00)
Qwen3-Max	Full	**2 / 1 / 89 / 0**	**100.00% (34.24, 100.00)**	**98.89% (93.97, 99.80)**	**66.67% (20.77, 93.85)**	**100.00% (95.86, 100.00)**
GPT-5	Base	2 / 22 / 68 / 0	100.00% (34.24, 100.00)	75.56% (65.75, 83.27)	8.33% (2.32, 25.85)	100.00% (94.65, 100.00)
GPT-5	Full	**2 / 7 / 83 / 0**	**100.00% (34.24, 100.00)**	**92.22% (84.81, 96.18)**	**22.22% (6.32, 54.74)**	**100.00% (95.58, 100.00)**

Supplementary Table S15 | Paired results for unaided and AI-assisted physician interpretation and reporting time

A shows confusion-matrix counts for both cohorts; B and C show prospective and multicenter screening performance, respectively; D shows paired transitions, accuracy differences and tests of case-level interpretation correctness; E shows report drafting and review time and paired time savings.

A. Both validation cohorts: confusion-matrix counts for unaided and AI-assisted physician interpretation

Cohort	Condition	n	Binary confusion matrix · Case counts			
			TP	FP	TN	FN
Prospective cohort	Unaided physician interpretation	1,458	291	35	1,083	49
Prospective cohort	Physician + AI	1,458	319	16	1,102	21
Multicenter cohort	Unaided physician interpretation	3,370	264	113	2,941	52
Multicenter cohort	Physician + AI	3,370	290	61	2,993	26

B. Prospective cohort: screening performance with unaided and AI-assisted interpretation (95% CI)

Metric	Unaided physician interpretation		AI-assisted interpretation	
	Unaided interpretation (95% CI)	n / N	Physician + AI (95% CI)	n / N
Sensitivity	85.59% (81.46, 88.92)	291 / 340	93.82% (90.74, 95.93)	319 / 340
Specificity	96.87% (95.68, 97.74)	1,083 / 1,118	98.57% (97.69, 99.12)	1,102 / 1,118
PPV	89.26% (85.43, 92.18)	291 / 326	95.22% (92.38, 97.04)	319 / 335
NPV	95.67% (94.32, 96.71)	1,083 / 1,132	98.13% (97.16, 98.77)	1,102 / 1,123
Accuracy	94.24% (92.92, 95.32)	1,374 / 1,458	97.46% (96.52, 98.15)	1,421 / 1,458
LR+	27.3393 (19.6751, 37.9891)	291 / 340	65.5592 (40.2745, 106.7180)	319 / 340
LR−	0.1488 (0.1148, 0.1928)	49 / 340	0.0627 (0.0414, 0.0948)	21 / 340

C. Multicenter cohort: screening performance with unaided and AI-assisted interpretation (95% CI)

Metric	Unaided physician interpretation		AI-assisted interpretation	
	Unaided interpretation (95% CI)	n / N	Physician + AI (95% CI)	n / N
Sensitivity	83.54% (79.06, 87.22)	264 / 316	91.77% (88.22, 94.32)	290 / 316
Specificity	96.30% (95.57, 96.91)	2,941 / 3,054	98.00% (97.44, 98.44)	2,993 / 3,054
PPV	70.03% (65.22, 74.43)	264 / 377	82.62% (78.31, 86.23)	290 / 351
NPV	98.26% (97.73, 98.67)	2,941 / 2,993	99.14% (98.74, 99.41)	2,993 / 3,019
Accuracy	95.10% (94.32, 95.78)	3,205 / 3,370	97.42% (96.83, 97.90)	3,283 / 3,370
LR+	22.5791 (18.7200, 27.2339)	264 / 316	45.9463 (35.7611, 59.0323)	290 / 316
LR−	0.1709 (0.1333, 0.2191)	52 / 316	0.0840 (0.0581, 0.1213)	26 / 316

D. Both validation cohorts: paired transitions in interpretation correctness, accuracy differences and statistical tests

Cohort	n	Paired classification counts				Paired difference	Statistical test
		a	b	c	d	Difference (95% CI)	P (exact)
Prospective cohort	1,458	1,371	3	50	34	0.032 (0.018, 0.046)	5.52E-12
Multicenter cohort	3,370	3,186	19	97	68	0.023 (0.013, 0.033)	8.63E-14

Errors in physician interpretation records include incorrect primary diagnoses and diagnostic omissions. Paired analyses use case-level correct or incorrect interpretation: a denotes correct interpretation under both conditions; b, correct unaided interpretation but incorrect AI-assisted interpretation; c, incorrect unaided interpretation but correct AI-assisted interpretation; and d, incorrect interpretation under both conditions. The accuracy difference is AI-assisted minus unaided interpretation, with a 95% confidence interval and an exact McNemar test P value.

E. Both validation cohorts: report drafting and review time and paired case-level time savings (min; 95% CI)

Cohort	Condition	Mean time or paired time savings, min (95% CI)
Prospective cohort	Unaided physician interpretation	10.498 (10.346, 10.650)
Prospective cohort	Physician + AI	5.733 (5.573, 5.893)
Prospective cohort	Paired time savings	4.766 (4.711, 4.821)
Multicenter cohort	Unaided physician interpretation	10.557 (10.452, 10.663)
Multicenter cohort	Physician + AI	5.809 (5.699, 5.920)
Multicenter cohort	Paired time savings	4.748 (4.711, 4.784)

Supplementary Table S16 | Quality of template and LLM-generated reports based on identical retrieved evidence

Differences are LLM report scores minus matched-evidence template scores. Means and differences are accompanied by 95% confidence intervals.

A. Prospective cohort: report-quality scores, paired differences and statistical tests

Rating dimension	n	Report-quality scores		Paired change	Significance tests	
		Template mean (95% CI)	LLM mean (95% CI)	Difference (95% CI)	P（Wilcoxon）	P (Holm)
Readability and clarity	1,458	0.797 (0.791, 0.802)	0.860 (0.855, 0.865)	0.063 (0.060, 0.066)	9.43E-190	1.89E-189
Clinical content completeness	1,458	0.853 (0.848, 0.859)	0.870 (0.865, 0.875)	0.016 (0.015, 0.018)	7.58E-82	7.58E-82

B. Multicenter cohort: report-quality scores, paired differences and statistical tests

Rating dimension	n	Report-quality scores		Paired change	Significance tests	
		Template mean (95% CI)	LLM mean (95% CI)	Difference (95% CI)	P（Wilcoxon）	P (Holm)
Readability and clarity	3,370	0.798 (0.794, 0.802)	0.862 (0.859, 0.866)	0.064 (0.062, 0.066)	1.00E-300	2.00E-300
Clinical content completeness	3,370	0.855 (0.852, 0.858)	0.871 (0.868, 0.875)	0.016 (0.015, 0.017)	3.15E-189	3.15E-189

Supplementary Table S17 | Access configurations and service sources for the six base-model backends

This table lists the model-access configurations used in this study. The four locally deployed open-weight models were accessed through the Data and Computing Platform, Hangzhou International Campus, Beihang University; Qwen3-Max and GPT-5 were accessed through test API services provided by the research team.

Model	Access configuration	Platform or service
Kimi-K2	Access to locally deployed model	Computing Center, Hangzhou International Campus, Beihang University
DeepSeek-V3.2	Access to locally deployed model	Computing Center, Hangzhou International Campus, Beihang University
GLM-4.5	Access to locally deployed model	Computing Center, Hangzhou International Campus, Beihang University
Llama-3.1-70B	Access to locally deployed model	Computing Center, Hangzhou International Campus, Beihang University
Qwen3-Max	Test API	Test API services provided by the research team
GPT-5	Test API	Test API services provided by the research team

Supplementary Table S18 | Positive and negative predictive values at different assumed prevalences

PPV and NPV were calculated at different assumed prevalences while holding source-cohort sensitivity and specificity point estimates constant. The 95% confidence intervals were obtained by monotonic transformation of the corresponding likelihood-ratio intervals (LR+ or LR−). This is a standardized analysis based on existing screening results, not a newly enrolled clinical cohort. Prevalence, PPV, NPV and their intervals are expressed as percentages. PPV and its intervals are reported to two decimal places; NPV and its intervals are reported to three decimal places.

Standardized predictive values based on prospective-cohort screening results

Model	Condition	Assumed prevalence	Standardized predictive values and 95% CI	
			PPV（95% CI）	NPV（95% CI）
Kimi-K2	Base	0.5%	1.40% (1.22, 1.61)	99.746% (99.709, 99.777)
Kimi-K2	Base	1%	2.78% (2.42, 3.19)	99.490% (99.417, 99.553)
Kimi-K2	Base	2%	5.46% (4.77, 6.24)	98.975% (98.830, 99.102)
Kimi-K2	Full	**0.5%**	**23.27% (15.90, 32.73)**	**99.958% (99.940, 99.971)**
Kimi-K2	Full	**1%**	**37.87% (27.53, 49.45)**	**99.916% (99.880, 99.941)**
Kimi-K2	Full	**2%**	**55.19% (43.43, 66.40)**	**99.830% (99.757, 99.880)**
DeepSeek-V3.2	Base	0.5%	0.49% (0.45, 0.52)	99.458% (99.337, 99.557)
DeepSeek-V3.2	Base	1%	0.97% (0.91, 1.05)	98.917% (98.676, 99.114)
DeepSeek-V3.2	Base	2%	1.95% (1.81, 2.09)	97.835% (97.360, 98.227)
DeepSeek-V3.2	Full	**0.5%**	**25.52% (17.15, 36.18)**	**99.957% (99.939, 99.969)**
DeepSeek-V3.2	Full	**1%**	**40.78% (29.39, 53.26)**	**99.913% (99.876, 99.938)**
DeepSeek-V3.2	Full	**2%**	**58.18% (45.68, 69.72)**	**99.824% (99.751, 99.876)**
GLM-4.5	Base	0.5%	1.15% (1.02, 1.30)	99.756% (99.717, 99.790)
GLM-4.5	Base	1%	2.29% (2.03, 2.58)	99.511% (99.433, 99.579)
GLM-4.5	Base	2%	4.52% (4.02, 5.08)	99.018% (98.861, 99.153)
GLM-4.5	Full	**0.5%**	**20.28% (14.13, 28.23)**	**99.952% (99.933, 99.965)**

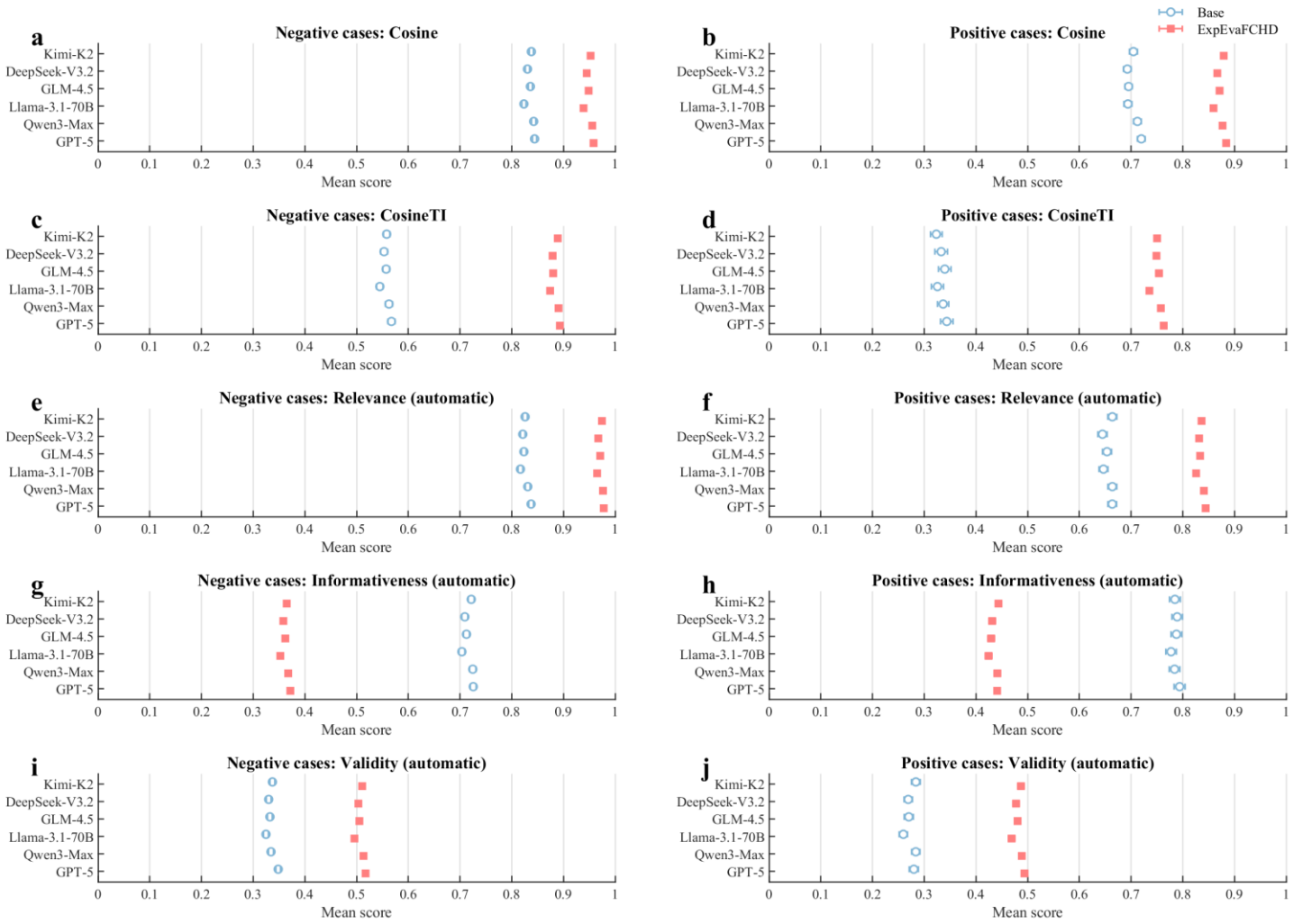
Model	Condition	Assumed prevalence	Standardized predictive values and 95% CI	
			PPV（95% CI）	NPV（95% CI）
GLM-4.5	Full	**1%**	**33.84% (24.86, 44.16)**	**99.903% (99.866, 99.930)**
GLM-4.5	Full	**2%**	**50.82% (40.06, 61.50)**	**99.805% (99.729, 99.860)**
Llama-3.1-70B	Base	0.5%	1.06% (0.93, 1.20)	99.721% (99.681, 99.756)
Llama-3.1-70B	Base	1%	2.11% (1.86, 2.39)	99.441% (99.360, 99.511)
Llama-3.1-70B	Base	2%	4.16% (3.68, 4.71)	98.876% (98.716, 99.017)
Llama-3.1-70B	Full	**0.5%**	**18.49% (13.02, 25.58)**	**99.943% (99.923, 99.958)**
Llama-3.1-70B	Full	**1%**	**31.32% (23.13, 40.86)**	**99.885% (99.845, 99.915)**
Llama-3.1-70B	Full	**2%**	**47.95% (37.81, 58.26)**	**99.768% (99.687, 99.828)**
Qwen3-Max	Base	0.5%	1.07% (0.93, 1.23)	99.689% (99.650, 99.724)
Qwen3-Max	Base	1%	2.13% (1.85, 2.45)	99.377% (99.299, 99.446)
Qwen3-Max	Base	2%	4.22% (3.67, 4.83)	98.749% (98.594, 98.887)
Qwen3-Max	Full	**0.5%**	**30.19% (19.75, 43.17)**	**99.961% (99.944, 99.973)**
Qwen3-Max	Full	**1%**	**46.50% (33.10, 60.43)**	**99.922% (99.887, 99.946)**
Qwen3-Max	Full	**2%**	**63.72% (49.99, 75.52)**	**99.842% (99.772, 99.891)**
GPT-5	Base	0.5%	1.73% (1.51, 1.97)	99.812% (99.778, 99.840)
GPT-5	Base	1%	3.41% (2.99, 3.89)	99.622% (99.555, 99.680)
GPT-5	Base	2%	6.66% (5.86, 7.55)	99.240% (99.105, 99.355)
GPT-5	Full	**0.5%**	**28.66% (18.95, 40.83)**	**99.964% (99.947, 99.976)**
GPT-5	Full	**1%**	**44.67% (31.97, 58.11)**	**99.928% (99.894, 99.951)**
GPT-5	Full	**2%**	**61.99% (48.70, 73.70)**	**99.854% (99.786, 99.901)**

Standardized predictive values based on multicenter-cohort screening results

Model	Condition	Assumed prevalence	Standardized predictive values and 95% CI	
			PPV（95% CI）	NPV（95% CI）
Kimi-K2	Base	0.5%	1.28% (1.14, 1.44)	99.723% (99.686, 99.757)
Kimi-K2	Base	1%	2.55% (2.27, 2.85)	99.446% (99.370, 99.512)
Kimi-K2	Base	2%	5.02% (4.49, 5.60)	98.886% (98.736, 99.019)
Kimi-K2	Full	**0.5%**	**16.31% (13.36, 19.78)**	**99.943% (99.922, 99.958)**
Kimi-K2	Full	**1%**	**28.15% (23.66, 33.14)**	**99.886% (99.844, 99.916)**
Kimi-K2	Full	**2%**	**44.19% (38.50, 50.03)**	**99.769% (99.685, 99.831)**
DeepSeek-V3.2	Base	0.5%	0.46% (0.43, 0.50)	99.379% (99.256, 99.481)
DeepSeek-V3.2	Base	1%	0.92% (0.86, 0.99)	98.759% (98.516, 98.962)

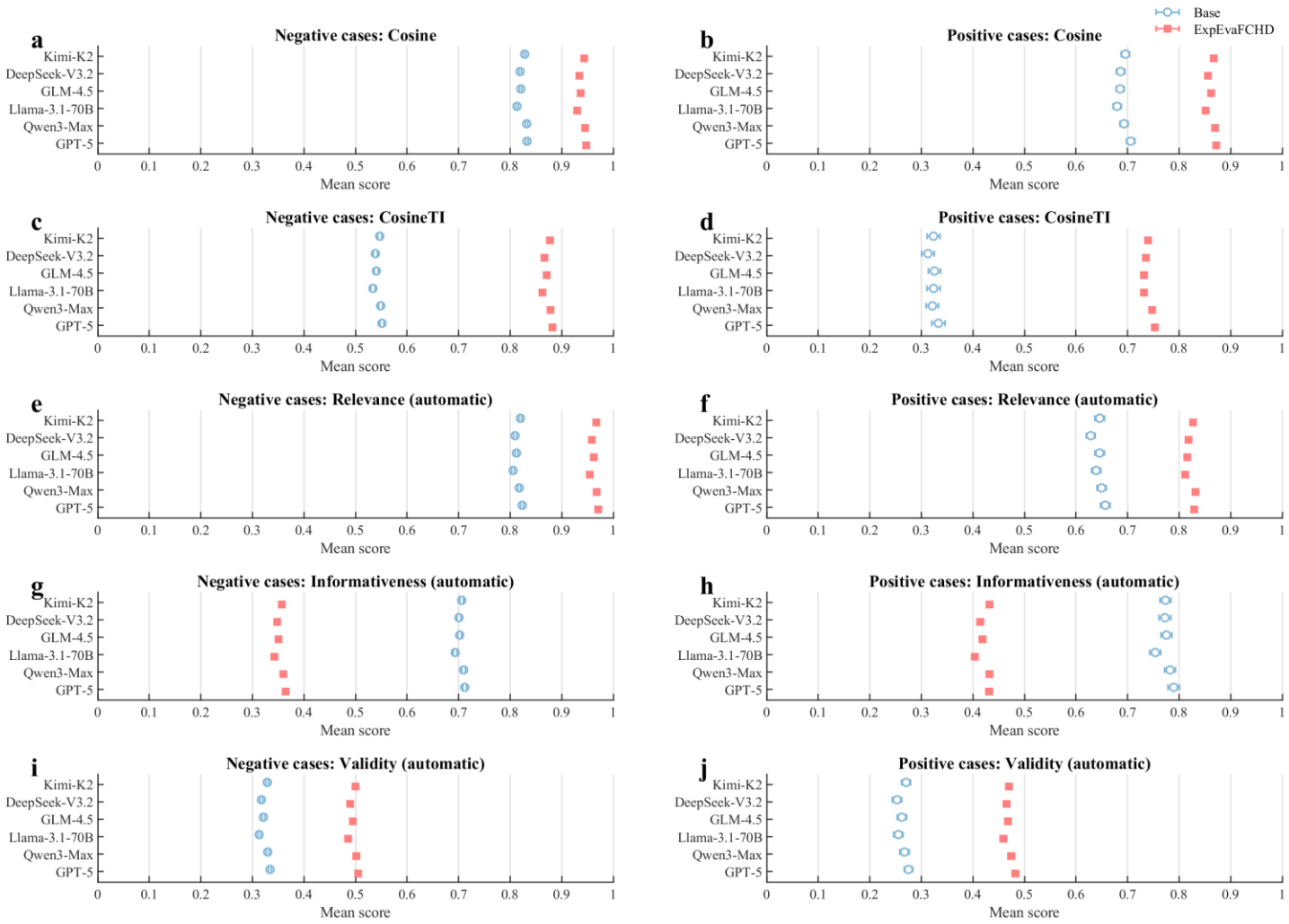
Model	Condition	Assumed prevalence	Standardized predictive values and 95% CI	
			PPV（95% CI）	NPV（95% CI）
DeepSeek-V3.2	Base	2%	1.85% (1.72, 1.99)	97.523% (97.047, 97.924)
DeepSeek-V3.2	Full	**0.5%**	**17.52% (14.25, 21.36)**	**99.942% (99.921, 99.957)**
DeepSeek-V3.2	Full	**1%**	**29.93% (25.04, 35.32)**	**99.883% (99.840, 99.914)**
DeepSeek-V3.2	Full	**2%**	**46.32% (40.30, 52.45)**	**99.763% (99.678, 99.826)**
GLM-4.5	Base	0.5%	1.07% (0.97, 1.18)	99.733% (99.692, 99.768)
GLM-4.5	Base	1%	2.13% (1.92, 2.35)	99.464% (99.382, 99.535)
GLM-4.5	Base	2%	4.21% (3.81, 4.64)	98.923% (98.760, 99.065)
GLM-4.5	Full	**0.5%**	**14.55% (12.01, 17.53)**	**99.936% (99.915, 99.953)**
GLM-4.5	Full	**1%**	**25.50% (21.52, 29.93)**	**99.872% (99.829, 99.905)**
GLM-4.5	Full	**2%**	**40.88% (35.66, 46.33)**	**99.742% (99.654, 99.808)**
Llama-3.1-70B	Base	0.5%	0.98% (0.88, 1.09)	99.696% (99.655, 99.733)
Llama-3.1-70B	Base	1%	1.94% (1.74, 2.17)	99.391% (99.308, 99.465)
Llama-3.1-70B	Base	2%	3.85% (3.46, 4.28)	98.778% (98.612, 98.924)
Llama-3.1-70B	Full	**0.5%**	**13.27% (11.01, 15.92)**	**99.926% (99.904, 99.944)**
Llama-3.1-70B	Full	**1%**	**23.53% (19.91, 27.57)**	**99.852% (99.806, 99.887)**
Llama-3.1-70B	Full	**2%**	**38.33% (33.43, 43.48)**	**99.702% (99.610, 99.772)**
Qwen3-Max	Base	0.5%	0.98% (0.87, 1.11)	99.666% (99.626, 99.702)
Qwen3-Max	Base	1%	1.95% (1.73, 2.21)	99.331% (99.251, 99.402)
Qwen3-Max	Base	2%	3.87% (3.43, 4.36)	98.657% (98.499, 98.799)
Qwen3-Max	Full	**0.5%**	**19.16% (15.48, 23.47)**	**99.947% (99.926, 99.961)**
Qwen3-Max	Full	**1%**	**32.26% (26.91, 38.13)**	**99.893% (99.852, 99.922)**
Qwen3-Max	Full	**2%**	**49.04% (42.65, 55.46)**	**99.783% (99.701, 99.843)**
GPT-5	Base	0.5%	1.60% (1.44, 1.77)	99.792% (99.756, 99.822)
GPT-5	Base	1%	3.16% (2.85, 3.49)	99.582% (99.511, 99.643)
GPT-5	Base	2%	6.18% (5.60, 6.81)	99.159% (99.016, 99.281)
GPT-5	Full	**0.5%**	**18.44% (14.96, 22.52)**	**99.948% (99.928, 99.963)**
GPT-5	Full	**1%**	**31.25% (26.12, 36.88)**	**99.896% (99.855, 99.925)**
GPT-5	Full	**2%**	**47.87% (41.67, 54.14)**	**99.790% (99.708, 99.848)**

Supplementary Fig. S1 | Five report-quality metrics for six models in the prospective cohort



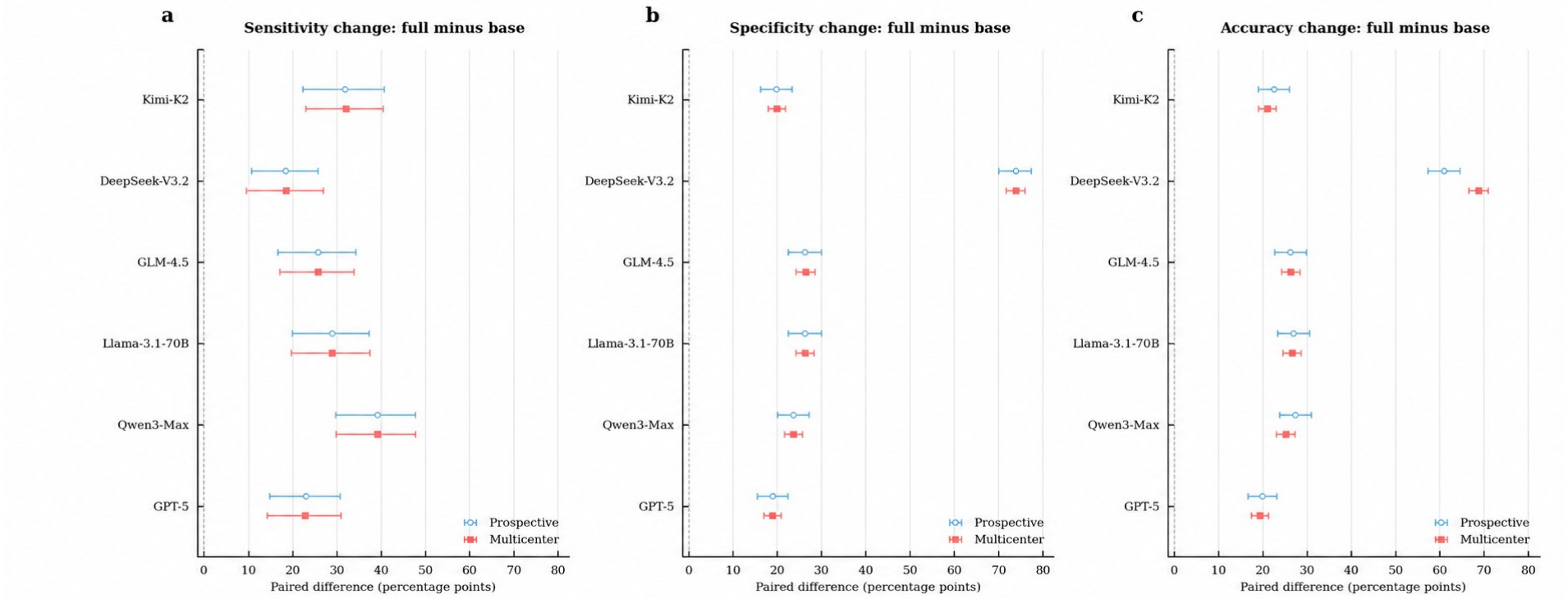
a,b, Cosine for negative and positive cases, respectively. c,d, CosineTI. e,f, Relevance. g,h, Informativeness. i,j, Validity. Blue open circles indicate Base; red squares indicate ExpEvaFCHD. Points are means, and error bars are 95% Student's t confidence intervals for the means. There were 1,118 negative and 340 positive cases. Detailed results are provided in **Supplementary Table S1**.

Supplementary Fig. S2 | Five report-quality metrics for six models in the multicenter cohort



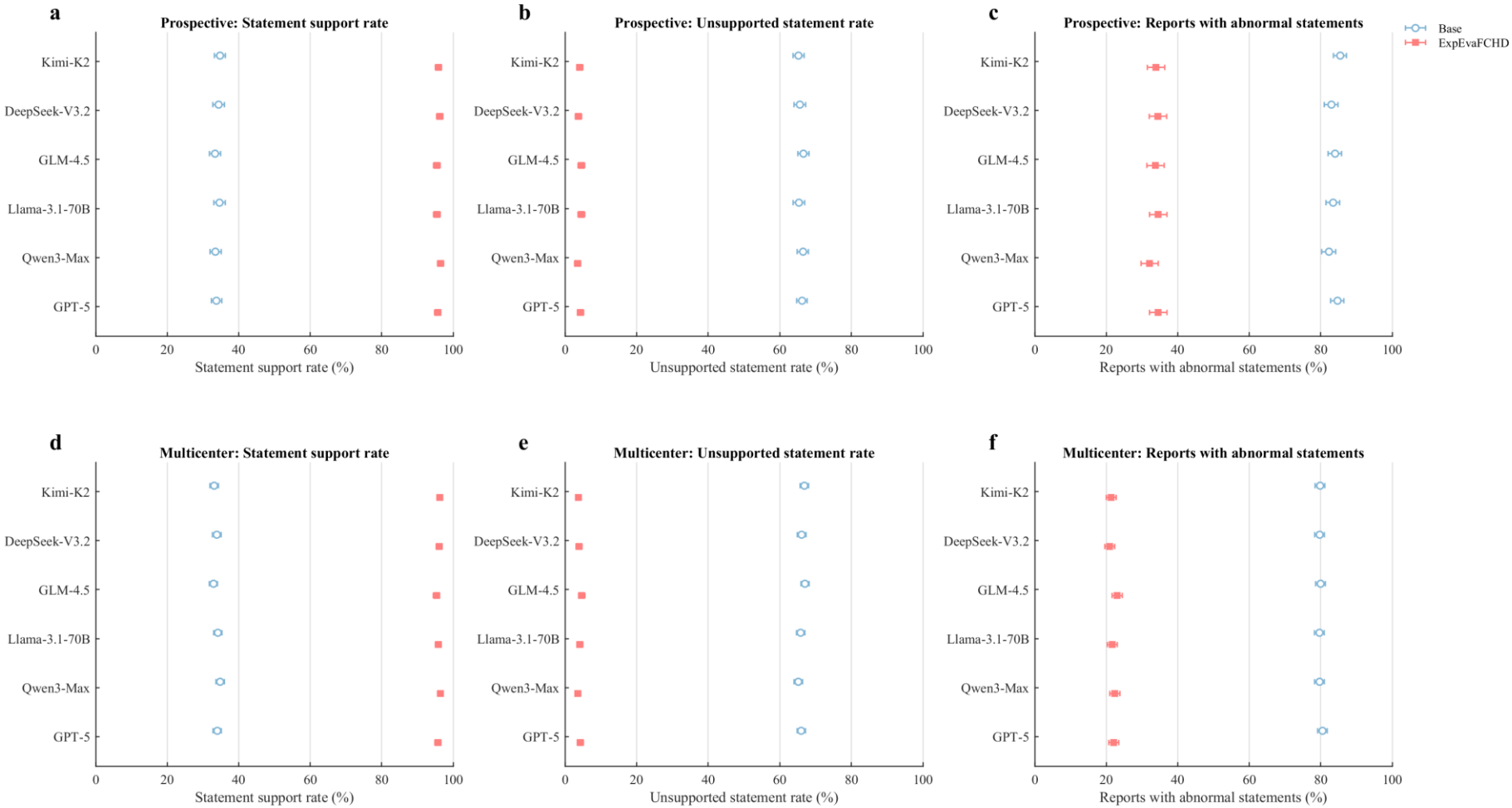
a,b, Cosine for negative and positive cases, respectively. c,d, CosineTI. e,f, Relevance. g,h, Informativeness. i,j, Validity. Blue open circles indicate Base; red squares indicate ExpEvaFCHD. Points are means, and error bars are 95% Student's t confidence intervals for the means. There were 3,054 negative and 316 positive cases. Detailed results are provided in **Supplementary Table S1**.

Supplementary Fig. S3 | Paired case-level screening improvements with the full system relative to base models



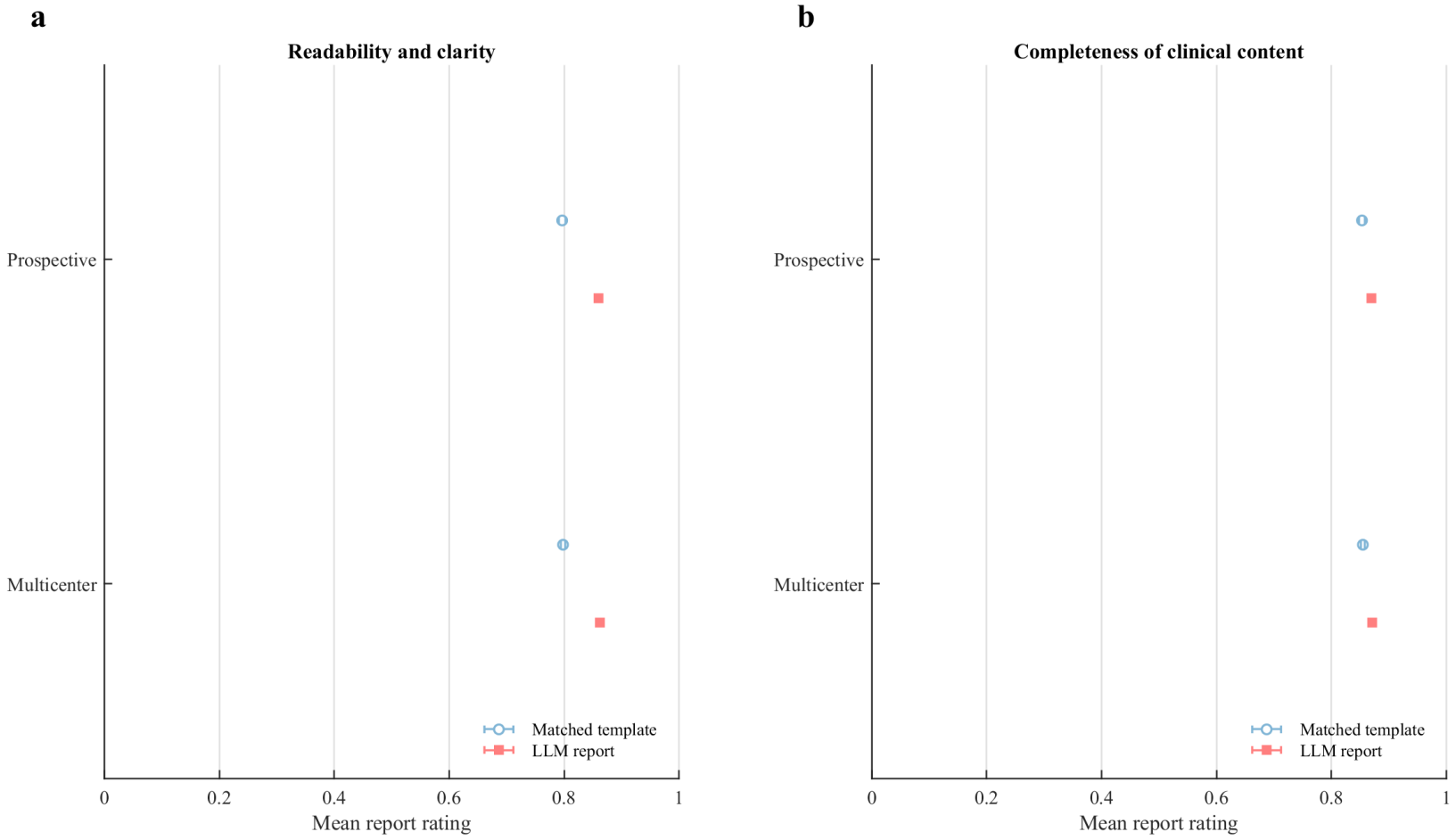
a, Sensitivity differences. b, Specificity differences. c, Accuracy differences. Differences are Full minus Base, expressed in percentage points. Blue open circles indicate the prospective cohort; red squares indicate the multicenter cohort. Error bars represent 95% confidence intervals constructed by combining the separate 97.5% Clopper–Pearson interval bounds for the two discordant-pair proportions. Detailed results are provided in **Supplementary Table S5**.

Supplementary Fig. S4 | Evidence support for abnormality statements and report coverage in model-generated reports



a–c, Statement support rates, unsupported-statement rates and the proportion of reports containing abnormality statements in the prospective cohort. d–f, Corresponding results in the multicenter cohort. Blue open circles indicate Base; red squares indicate ExpEvaFCHD. Error bars represent 95% confidence intervals. The first two measures use the number of abnormality statements as the denominator; the report proportion uses all reports. Detailed results are provided in **Supplementary Table S11**. Physicians audited abnormality statements in model-generated reports during daily final review.

Supplementary Fig. S5 | Quality scores for template and LLM-generated reports using identical retrieved evidence



a, Readability and clarity. b, Clinical content completeness. Blue open circles indicate matched-evidence templates; red squares indicate LLM-generated reports. Points are means, and error bars represent 95% confidence intervals. Results are shown separately for the prospective and multicenter cohorts. Detailed results are provided in **Supplementary Table S16**.

Supplementary Fig. S6 | Geographical distribution of the expert-database center and multicenter validation hospitals



Establishment and Testing of Data: Beijing
Beijing Anzhen Hospital
Multicenter Validation (Northeast): Liaoning
Shenyang Women's and Children's Hospital Huludao Maternal and Child Health Hospital
Multicenter Validation (Northwest): Ningxia
Ningxia Women and Children's Hospital Guyuan Maternal and Child Health Hospital
Multicenter Validation (Center): Anhui
Anhui Women and Children's Medical Center Huaibei People's Hospital
Multicenter Validation (Southeast): Fujian
Fujian Provincial Maternal and Child Health Hospital Fujian Provincial Obstetrics and Gynecology Hospital Fuzhou Second General Maternity and Child Health Care Hospital Jianyang District Maternal and Child Health Hospital The Second Hospital of Sanming City Zhangpu County Maternal and Child Health Hospital
Multicenter Validation (Southwest): Guangdong
Guangzhou Women and Children's Medical Center Guangdong Provincial People's Hospital Gaozhou Maternal and Child Health Hospital

a, Beijing was the center for FetalCardioDB construction and prospective validation; Fujian, Ningxia, Guangdong, Anhui and Liaoning were the multicenter external-validation regions. b, The 15 participating hospitals across five provinces.